MATH 202A: Introduction to Topology and Analysis

By William Nguyen

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8/25/26 Lecture 1: Introduction

Definition 1.1: Metric Spaces

Given a set X, $d : X \times X \to \mathbb{R}$ is a *metric* if

1. $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) \leq d(x, y) + d(y, z)$

We then say (X, d) is a *metric space*

Lets start with $\mathbb{Q}$ with it's metric being $d\left(\frac{m_1}{n_1}, \frac{m_2}{n_2}\right) = \left|\frac{m_1}{n_1} - \frac{m_2}{n_2}\right|$

Construction of the Reals

First, why do we want $\mathbb{R}$? The answer is that in $\mathbb{Q}$, we don't have solutions to many polynomial equations, and that sets don't have well defined suprema and infima.

Definition 1.2: Cauchy Sequences

We say a sequence $\{x_n\} \in X$ is *Cauchy* if $\forall \epsilon > 0$, $\exists N > 0$ such that $\forall n, m > N$,

$$d(x_n, x_m) < \epsilon$$

Now, the goal is to define real numbers as Cauchy sequences of $\mathbb{Q}$, but right now, we have a problem with unique representation. There are many Cauchy sequences that approach the same limit, so we can fix that.

Let $X' = \{\{x_n\} \in \mathbb{Q} : \{x_n\} \text{ is Cauchy}\}$, and define the following equivalence relation. Say

$$\{x_n\} \sim \{y_n\} \text{ if, and only if } d(x_n, y_n) \to 0 \text{ as } n \to \infty$$

With this, we can define $\mathbb{R}$ to be equivalence classes of X'. Note that with this definition, we can define a natural metric on $\mathbb{R}$ by

$$d_{\mathbb{R}}([x_n], [y_n]) = \lim_{n \to \infty} d_{\mathbb{Q}}(x_n, y_n)$$

and this is well defined (choice of representative is irrelevant).

Definition 1.3: Completeness

We say a metric space X is *complete* if all Cauchy sequences converge.

Theorem 1.4: Completeness of $\mathbb{R}$

With $d(x, y) = |x - y|$, $\mathbb{R}$ is complete.

Here are some interesting examples of metric spaces.

- Given a connected graph, let $d(x, y)$ be the length of the shortest path between the vertices x and y.

- Given a set of strings of length n from an alphabet, let $d(w_1, w_2)$ be the number of entries in which w_1 and w_2 differ. This is called the *Hamming distance*.
- Consider $\mathbb{Q}$, and for $x, y \in \mathbb{Q}$, write $|x - y| = 2^k \frac{a}{b}$ where a and b are odd. This representation is unique, so define $d(x, y) = 2^{-k}$. This is called the 2-adic metric. It turns out by Ostrowski's theorem, every nontrivial absolute value on $\mathbb{Q}$ is either the normal one, or a p-adic one, which is cool!
- Consider $X = \{\text{functions on } [0, 1]\}$ with $d(f, g) = \sup |f(x) - g(x)|$. This is well defined since $[0, 1]$ is compact.

Definition 1.5: Completions

Given a metric space (X, d), we say that (X^*, d^*) is a *completion* of (X, d) if it is complete and there exists an injective function $f : X \to X^*$ such that

$$d^*(f(x), f(y)) = d(x, y)$$

and also that $f(X)$ is dense in X^*. By dense, we mean that for any $x^* \in X^*$, there is a sequence of points in X that approach it.

With this definition, we see that $\mathbb{R}$ is a completion of $\mathbb{Q}$, and we take $f(x) = \{x_n = x\}$ to be the injective function. A quick question one may ask is why we need density? The reason is that we want uniqueness. For example, we don't want to say that $\mathbb{R}$ with an additional far point off the real line is a completion of $\mathbb{Q}$.

Theorem 1.6: Existence of Completion

Every metric space has a completion

Proof Sketch: Take the desired set to be equivalence classes of Cauchy sequences in X, and show that this is a completion.

Now, we also want to show that completions of metric spaces are unique, but we need a notion of equivalence first.

Definition 1.7: Isometries

A map $f : X \to Y$ of metric spaces is an *isometry* if

$$d_X(x, y) = d_Y(f(x), d(f(y)))$$

An *isomorphism* is a bijective isometry.

Theorem 1.8: Uniqueness of Metric Spaces

A completion of a metric space is unique up to isomorphism.

An example of such an isomorphism is letting $X = \mathbb{R}$, $Y = \{(x, 0) \in \mathbb{R}^2\}$, and $f(x) = (x, 0)$, where X and Y have the Euclidean metric. It can be easily checked that f is an isomorphism.

8/28/26 Lecture 2: Introduction to Measure

We want to assign a size to subsets $A \subseteq \mathbb{R}$. If $A = [a, b]$, it's quite natural to say its size is $b - a$. Also, we would want the size of $\cup I_n$ with I_i are disjoint to be the sum of the sizes of I_n.

Digression on Probability Theory

We can model a sequence of coin flips as an element if $\{T, H\}^{\mathbb{N}}$. However, here is another way to do it. For $x \in [0, 1]$, write x in binary like $x = (0.1001\ldots)_2$ for instance. Then, associate coin tosses to points in $[0, 1]$ by mapping $T \to 0$ and $H \to 1$, so that $HTTH\ldots \to (0.1001\ldots)_2$. This allows us to interpret statements about probability as statements about sizes of subsets of $[0, 1]$.

For instance,

$$\begin{aligned}\{x \in \{T, H\}^{\mathbb{N}} : x \text{ begins with an } H\} &= \{x \in [0, 1] : x = 0.1\ldots\} \\ &= [1/2, 1]\end{aligned}$$

Thus, one can conclude that the probability that a sequence of flips starts with an H is $1/2$.

Here's another question, one that is harder to solve: How do we show that the number of head and tails get closer and closer to being equal with probability 1? Let's try to represent this set of sequences as a subset of $[0, 1]$. It is clear that the set of such sequences is in correspondence with

$$\mathcal{N} = \left\{ x \in [0, 1] : \lim_{n\to\infty} \frac{1}{N} \sum_{i=1}^{N} b_n(x) = 1/2 \right\}$$

where b_n is the nth binary digit of x. However, this has limits and, so is there a way to make this more digestible? The answer is yes. Consider

$$\bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ x \in [0, 1] : \left| \frac{1}{n} \sum_{i=1}^{n} b_i(x) - 1/2 \right| < 1/k \right\}$$

Parsing what this set is, we see it says that belonging in this set means that "for all k, there exists an N, such that for all $n \geq N$, $\left|\frac{1}{n}\sum_{i=1}^{n} b_i(x) - 1/2\right| = 1/k$". That is, we just turned the notion of a limit into the intersection and union of simpler sets! Soon, we will see this technique makes it much easier to measure sets with properties like limits approaching a certain value. Before that, some topology.

Definition 2.1: Negligible Sets

A set A is *negligible* if, $\forall \epsilon > 0$, there is a countable collection of (open?) intervals $\{I_n\}_{n=1}^{\infty}$ such that

$$A \subseteq \bigcup_{n=1}^{\infty} I_n$$

and $\sum_{n=1}^{\infty} |I_n| < \epsilon$.

The following is a theorem we won't prove, but we will state it here.

Theorem 2.2: Borel's Normal Number Theorem

The set $\mathcal{N}^c = [0,1]\backslash\mathcal{N}$ is negligible.

Thus, we can conclude that the probability that the number of heads and tails aren't equal in the long run occurs with probability 0. Moving past probability, let's explore some other useful results about this topological/measure theoretic term we just defined.

Proposition 2.3

If A is countable, then it is negligible.

Proof: Let $\epsilon > 0$ and $A = \{a_1, a_2, \ldots\}$. Then, letting $I_n = (a_n - \epsilon/2^{n+2}, a_n + \epsilon/2^{n+2})$, we see that $A \subseteq \cup_{n=1}^{\infty} I_n$. Then, as $|I_n| = \epsilon/2^{n+1}$, we see

$$\sum_{b=1}^{\infty} |I_n| = \sum_{n=1}^{\infty} \epsilon/2^{n+1} = \epsilon/2 < \epsilon$$

Thus, A is countable.

■

Proposition 2.4

Let $\{A_n\}_{n=1}^{\infty}$ be a collection of negligible sets. Then $\cup_{n=1}^{\infty} A_n$ is also negligible.

Proof: Let $\epsilon > 0$. For each A_i, we can find a collection of intervals $\{I_{n,i}\}$ such that $A_i \subseteq \cup_{n=1}^{\infty} I_{n,i}$ and $\sum_{n=1}^{\infty} |I_{n,i}| < \epsilon/2^i$.

Now, consider the set $\{I_{n,i} : i, n \in \mathbb{N}\}$. We see that this is a countable collection of intervals, $\cup_{i=1}^{\infty} \subseteq \cup_{n,i\geq 1} I_{n,i}$, and

$$\begin{aligned}\sum_{n,i\geq 1} |I_{n,i}| &= \sum_{i=1}^{\infty}\sum_{n=1}^{\infty} |I_{n,i}| \\ &< \sum_{i=1}^{\infty} \epsilon/2^i \\ &= \epsilon\end{aligned}$$

Thus, $\cup_{n=1}^{\infty} A_n$ is negligible.

■

Now, small contingency. Is it possible that we can have a $\mathcal{N} \subseteq [0,1]$ with both $\mathcal{N}^c = [0,1]\backslash\mathcal{N}$ and $\mathcal{N}$ being negligible? Do non-negligible sets even exist? Our intuition says this shouldn't occur, but it is a suprisingly nontrivial result to show!

Proposition 2.5

$[0,1]$ is not negligible.

Before we prove this result however, we need some more tools!

Definition 2.6: Open Covers

Let $C \subseteq \mathbb{R}$. An *open cover* $\mathcal{O} = \{\mathcal{U}_\alpha\}_{\alpha \in I}$ is a collection of open sets such that $C \subseteq \cup_{\alpha \in I} \mathcal{U}_\alpha$.

Definition 2.7: Compactness

Let $C \subseteq \mathbb{R}$. We say that C is *compact* if every open cover of C has a finite subcover. That is, given an open cover $C \subseteq \cup_{\alpha \in I} \mathcal{U}_\alpha$, then we can write $C \subseteq \cup_{i=1}^n \mathcal{U}_{\alpha_1}$ for some $\mathcal{U}_{\alpha_i} \in \{\mathcal{U}_\alpha\}$.

Theorem 2.8: Heine-Borel Theorem

A set $C \subseteq \mathbb{R}$ is compact if, and only if, C is closed and bounded.

Now, let's try to prove Proposition 2.5 To do this, we will prove the following lemma. The full proof will be shown next lecture, but here is the proof sketch.

Lemma 2.9

If $\mathcal{O} = \{I_\alpha\}_{\alpha \in J}$ is a collection of open intervals such that $[a,b] \subseteq \cup_\alpha I_\alpha$, then $\sum_\alpha |I_\alpha| > b - a$.

Proof Sketch: Given an open cover (of intervals), pass to a finite cover. Thus, we only need to show this result for finite collections. To do this, we will use induction. For $k = 1$, we see that if $[a,b] \subseteq [c,d]$, indeed $d - c \geq b - a$. For the inductive step, wlog, write $[a,b] = [0,1]$. Then, considering a cover of $k+1$ intervals, note that one interval must contain the endpoint 1. Let this interval be I^*. Then, apply the inductive step to the interval $[0,1]\backslash I^*$ as this is covered with k intervals.

8/31/26 Lecture 3: Introduction to Measure

Proposition 3.1

A set $\mathcal{U} \subseteq \mathbb{R}$ is open if $\mathcal{U}$ can be written as a union of countable many disjoint open intervals.

Theorem 3.2: Existence of Non-Negligible Sets

The interval $[0,1]$ is not negligible

Before we prove this theorem, we need a short lemma.

Lemma 3.3

If $\{I_n\}_{n=1}^N$ is a finite collection so that $[a,b] \subseteq \bigcup_{i=1}^N I_i$, then $[a,b] \leq \sum_{i=1}^N |I_i|$.

Proof: We will use induction on N, the size of the collection. First, for the $N = 1$, we see clearly that if $[a,b] \subseteq (c,d)$, then $d - c > b - a$.

Now, suppose it holds for $N-1$, and suppose $\{I_n\}_{n=1}^N$ cover $[a,b]$. Then, there must be an n^* so that $b \in I_{n^*}$. Now, if $I_{n^*} = (c,d)$, then we see that $\{I_n\}_{n=1}^{N-1}$ must cover $[a,c] = [a,b] \backslash I_{n^*}$.

By the inductive step, we know that $\sum_{i=1}^{N-1} |I_i| \geq c - a$. Thus, we see that

$$\sum_{i=1}^{N} |I_i| \geq c - a + (d - c) = d - a \geq b - a$$

which was to be proved.

■

Now, we proceed with the proof of Theorem 3.2.

Proof of Theorem 3.2: Suppose that $[0,1] \subseteq \cup_{n=1}^{\infty} I_n$. By compactness, there exists a finite subcollection of the I_i such that $[0,1] \subseteq \cup_{n=1}^{N} I_{n_k}$. Then, by our proposition, we know that $\sum_{k=1}^{N} |I_{n_k}| \geq 1$, so clearly $\sum_{n=1}^{\infty} |I_n| \geq 1$. Thus, all interval coverings have combined length greater than or equal to 1, so it cannot be epsilon close to 0.

■

Now, moving past these topological notions, we want to try to construct a function λ on some subsets $A \subseteq \mathbb{R}$ such that $\lambda(A)$ assigns a size to A. So far, we want that if $\mathcal{U} \subseteq \mathbb{R}$ and $\mathcal{U} = \cup_{n=1}^{\infty} I_n$ with I_n being disjoint intervals, then we should have $\lambda(\mathcal{U}) = \sum_n |I_n|$.

Similarly, if we have an $A \subseteq [0,1]$ that is closed, then as $A \backslash [0,1] = \mathcal{U}$ is open, then we should have that $\lambda(A) = 1 - \lambda(\mathcal{U})$.

Here is a first attempt: For $A \in \mathcal{P}(\mathbb{R})$ define

$$\begin{aligned}\lambda^*(A) &= \inf\left\{\sum_{n=1}^{\infty} |I_n| \ : \ A \subseteq \bigcup_{n=1}^{\infty} I_n\right\} \\ &= \inf\{\lambda(\mathcal{U}) \ : \ \mathcal{U} \text{ is open and } A \subseteq \mathcal{U}\}\end{aligned}$$

This is actually not a bad attempt, and motivates the following definition!

Definition 3.4: Outer Measures

For a set X, a function $\mu^* : \mathcal{P}(X) \to [0,1]$ is an *outer measure* if

1. $\mu^*(\emptyset) = 0$
2. $\mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$, $A_n \subseteq X$
3. If $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$

Proposition 3.5

λ^*, as defined above, is an outer measure.

Proof: 1 and 3 follow immediately from the definition of λ^*, so we just need to prove 2. Start by letting $\{A_n\} \subseteq \mathbb{R}$ and let $\epsilon > 0$. Then, for each A_n, pick a collection of intervals $\{I_{n,j}\}$ so that $A_n \subseteq \cup_j I_{n,j}$ and that $\sum_j |I_{n,j}| \leq \lambda^*(A_n) + \epsilon/2^n$ (we can do this by the definition of an infimum). Then, considering the countable collection $\{I_{n,j}\}$ we see that $\cup_n A_n \subseteq \cup_{n,j} I_{n,j}$ and that

$$\begin{aligned}\lambda^*\left(\bigcup_n A_n\right) &\leq \sum_{n,j} |I_{n,j}| \\ &= \sum_n \left(\sum_j |I_{n,j}|\right) \\ &= \sum_n \lambda^*(A_n) + \epsilon\end{aligned}$$

Thus, $\lambda^*\left(\bigcup_n A_n\right) \leq \sum_n \lambda^*(A_n) + \epsilon$ for all ϵ. Thus, $\lambda^*\left(\bigcup_n A_n\right) \leq \sum_n \lambda^*(A_n)$. ■

Now, we will come back to outer measures soon, but first, we should define what a measure is. To do that, we need a suitable class of subsets of $\mathbb{R}$ on which a measure can exist.

Definition 3.6: σ-Algebras

For a set X, a subset $\mathcal{A} \subseteq \mathcal{P}(X)$ is a *σ-algebra* if

1. $\emptyset, X \in \mathcal{A}$
2. If $A_n \in \mathcal{A}$, then $\cup_n A_n \in \mathcal{A}$
3. If $A \in \mathcal{A}$ then $A^c = X \backslash A \in \mathcal{A}$

Corollary 3.7

If $A_n \in \mathcal{A}$, then $\cap_n A_n \in \mathcal{A}$

Proof: This follows immediately from applying De Morgan's laws to the definition of a σ-algebra. ■

Some examples of σ-algebras on any X are $\{\emptyset, X\}$ and $\mathcal{P}(X)$. With this definition, we can now define a measure.

Definition 3.7: Measures and Measure Spaces

A *measure space* is a triple $(X, \mathcal{A}, \mu)$ where $\mathcal{A}$ is a σ-algebra and $\mu : \mathcal{A} \to [0, \infty]$ is a map that satisfies the following properties:

1. $\mu(\emptyset) = 0$
2. If $A_n \in \mathcal{A}$ is a countable collection of pairwise disjoint sets, then $\mu(\cup_n A_n) = \sum_n \mu(A_n)$

We call μ a *measure*.

9/2/26 Lecture 4: Measures and σ-Algebras

Today, we will construct a measure on a σ-algebra using an outer measure.

Examples

1. For any set X, let $\mathcal{A} = \mathcal{P}(X)$. Then, we can let $\mu(A) = \#A$, where $\#A$ is the cardinality of A. We call μ the counting measure
2. Let X and $\mathcal{A} = \mathcal{P}(X)$. Then, fix $x \in X$, and define

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & \text{otherwise} \end{cases}$$

   Then, $\delta_x(A)$ is a measure, called a point mass.

Proposition 4.1

Let $(X, \mathcal{A}, \mu)$ be a measure space. Then, the following hold.

1. Let $\{A_n\}_{n=1}^N$ be a finite subset of $\mathcal{A}$ that are pairwise disjoint. Then, we have

$$\mu\left(\bigcup_{n=1}^N A_n\right) = \sum_{n=1}^N \mu(A_n)$$

2. If $A, B \in \mathcal{A}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$
3. If $A_n \in \mathcal{A}$ for each $n \in \mathbb{N}$ not necessarily disjoint, then

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) \leq \sum_{n=1}^\infty \mu(A_n)$$

4. If $A_1 \subseteq A_2 \subseteq \cdots$ are an increasing sequence of sets in $\mathcal{A}$, then

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

5. If $A_1 \supseteq A_2 \supseteq \cdots$ is a decreasing sequence in $\mathcal{A}$ and $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^\infty A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

Proof:
1) Given $A_1, \ldots, A_N$, set $A_m = \emptyset$ for $m > N$. Then, we see that

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{n=1}^\infty \mu(A_n) = \sum_{n=1}^N \mu(A_n)$$

and as $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{N} A_n$, we have our desired result.

2) Since $B = A \cup (B \backslash A)$, then we see that

$$\mu(B) = \mu(A) + \mu(B \backslash A)$$

as A and $B \backslash A$ are disjoint. Then, as $\mu(B \backslash A) \geq 0$, we have that $\mu(B) \geq \mu(A)$.

3) Define $B_1 = A_1$, $B_2 = A_2 \backslash A_1$ and $B_3 = A_3 \backslash (A_1 \cup A_2)$. In general, let $B_n = A_n \backslash \left(\bigcup_{i=1}^{n-1} A_i \right)$. Now, we see that the B_i are disjoint, so

$$\mu\left(\bigcup_{n=1}^{\infty} A_n \right) = \mu\left(\bigcup_{n=1}^{\infty} B_n \right) = \sum_{n=1}^{\infty} \mu(B_n)$$

Then, since $\mu(B_n) \leq \mu(A_n)$ as $B_n \subseteq A_n$, we see that

$$\mu\left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

4) This is like 5 so we will do 5.

5) Let $C_n = A_1 \backslash A_n$. By 4, since C_n is increasing, we see that

$$\mu\left(\bigcup_{n=1}^{\infty} C_n \right) = \lim_{n \to \infty} \mu(C_n) = \lim_{n \to \infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_{n \to \infty} (\mu(A_n))$$

However, we see that

$$\bigcup_{n=1}^{\infty} C_n = A_1 \backslash \bigcap_{n=1}^{\infty} A_n$$

so

$$\mu\left(\bigcup_{n=1}^{\infty} C_n \right) = \mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n \right)$$

Thus, we have our desired equality.

■

Note that the condition that $\mu(A_1)$ has finite measure is necessary in 5. This can be seen by the following counterexample. Take the counting measure on $\mathbb{R}$, and consider the sequence of decreasing sets

$$A_n = [n, \infty)$$

Then, we see that $\mu(A_n) = \infty$ for each n, but $\bigcap_n A_n = \emptyset$.

Definition 4.2: σ-algebra Generated by a Collection

Let $\mathcal{C}$ be a collection of subsets of a set X. Then, the *σ-algebra generated by $\mathcal{C}$* is

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra containing } \mathcal{C}\}$$

Since the intersection of σ-algebras is a σ-algebra, $\sigma(\mathcal{C})$ is indeed a σ-algebra.

In a sense, we say that $\sigma(\mathcal{C})$ is the smallest σ-algebra containing $\mathcal{C}$ (since any σ-algebra containing $\mathcal{C}$ must contain $\sigma(\mathcal{C})$).

Definition 4.3: Borel σ-algebra

Let X be a topological space, and let G be it's set of open sets. Then, the *Borel σ-algebra* is

$$\mathcal{B} := \sigma(G)$$

We call $A \in \mathcal{B}$ a *Borel set*.

Proposition 4.4

Consider $\mathbb{R}$ with the normal topology, and let $\mathcal{B}$ be the Borel σ-algebra. Then, $\mathcal{B}$ is generated by each of the following collections of subsets.

1. $\mathcal{C}_1 = \{(a,b) : a,b \in \mathbb{R}\}$
2. $\mathcal{C}_2 = \{[a,b] : a,b \in \mathbb{R}\}$
3. $\mathcal{C}_3 = \{(a,b] : a,b \in \mathbb{R}\}$
4. $\mathcal{C}_4 = \{(a,\infty) : a \in \mathbb{R}\}$

Proof: We won't do all of them, but we will show a few. The idea is to show that we can either generate the open sets with this collection, or the other way around.

1) First, any open set can be written as $\cup_{n=1}^{\infty} I_n$ where $I_n = (a,b), (-\infty, a)$ or (a,∞). Now, we can write

$$(a,\infty) = \cup_n (a, a+n)$$

and similarly,

$$(-\infty, a) = \cup_n (a-n, a)$$

Thus, $(a,b), (-\infty,a)$ and (a,∞) are in $\sigma(\mathcal{C}_1)$. Thus, we see that $\mathcal{B} \subseteq \sigma(\mathcal{C}_1)$. Since $C_1 \subseteq \mathcal{B}$, we equality.

2) The goal is to show that $\sigma(\mathcal{C}_2) \subseteq \mathcal{B}$ and that $\mathcal{B} \subseteq \sigma(\mathcal{C}_2)$. Since

$$[a,b] = \bigcap_n (a - 1/n, b + 1/n)$$

we see that $\mathcal{C}_2 \subseteq \sigma(\mathcal{C}_1) = \mathcal{B}$. Thus, $\sigma(\mathcal{C}_2) \subseteq \mathcal{B}$. Then, as

$$(a,b) = \bigcup_n [a + 1/n, b - 1/n]$$

we see that $\mathcal{C}_1 \subseteq \sigma(\mathcal{C}_2)$, so $\mathcal{B} = \sigma(\mathcal{C}_1) \subseteq \sigma(\mathcal{C}_2)$. Thus, we have equality.

3) and **4)** are very similar, so we leave the proof to the reader.

■

Definition 4.5: μ^*-Measurability

For an outer measure μ^* on a set X, we sey that $A \subseteq X$ is μ^*-*measurable* if, for any $E \subseteq X$, then

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

9/4/26 Lecture 5: More on Measures

Now, given an outer measure μ^* and the definition of μ^*-measurable sets, it turns out we can construct a σ-algebra and define a measure μ on it!

Theorem 5.1: Construction of Measures from Outer Measures

Given an outer measure μ^*, the collection of μ^*-measurable sets, denoted $\mathcal{A}$, is a σ-algebra, and the restriction of μ^* to $\mathcal{A}$ is a measure. Additionally, $\mathcal{A}$ contains all the null sets of μ^*

Proof: First, we show that $\mathcal{A}$ is indeed a σ-algebra. Firstly, from the symmetry of the definition of μ^*-measurability, we see that if $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$.

Now, we want to show that finite unions of sets in $\mathcal{A}$ are in $\mathcal{A}$. Firstly, notice that, in order to show that $A \in \mathcal{A}$, all we need is that for all $E \subseteq X$, $\mu^*(E) \supseteq \mu^*(E \cap A) + \mu^*(E \cap A^c)$, as the opposite inequality follows from subadditivity.

Let $A, B \in \mathcal{A}$. We want to show that $A \cup B \in \mathcal{A}$. First, for $E \subseteq X$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Then, applying the μ^*-measurability of B to the set $E \cap A^c$, we see taht

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) \\
&= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap (A \cup B)^c) \\
&\geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)
\end{aligned}$$

where the last inequality comes from applying subadditivity to the fact that $A \cup (A^c \cap B) = A \cup B$. Thus, we see that $A \cup B$ is μ^*-measurable, and by induction, we see that all finite unions of μ^*-measurable sets is μ^*-measurable.

Now, if $A_n \in \mathcal{A}$, we want to show that $\cup_n A_n \in \mathcal{A}$. Notice that it suffices to assume that A_n are disjoint as we can write

$$B_1 = A_1, \quad B_2 = A_1 \backslash A_2, \cdots \quad B_n = A_n \backslash \bigcup_{i=1}^{n-1} A_i$$

and see that the B_i are disjoint, $\cup_{i=1}^{\infty} B_i = \cup_{i=1}^{\infty} A_i$, $B_i \in \mathcal{A}$, and $\mu^*(B_i) \leq \mu^*(A_i)$. Thus, assuming that the A_i are disjoint, we have

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap A_1) + \mu^*(E \cap A_1^c) \\
&= \mu^*(E \cap A_1) + \mu^*(E \cap A_1^c \cap A_2) + \mu^*(E \cap A_1^c \cap A_2^c) \\
&= \mu^*(E \cap A_1) + \mu^*(E \cap A_2) + \mu^*(E \cap (A_1 \cup A_2)^c) \\
&\vdots \\
&= \sum_{i=1}^{N} \mu^*(E \cap A_i) + \mu^*\left(E \cap \left(\bigcup_{i=1}^{N} A_i\right)^c\right) \\
&\geq \sum_{i=1}^{N} \mu^*(E \cap A_i) + \mu^*\left(E \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right)
\end{aligned}$$

for all N, where the last step holds from the fact that $\left(\bigcup_{i=1}^N A_i\right)^c \supseteq (\bigcup_{i=1}^\infty A_i)^c$. Then, as this holds for all N, we can pass to the limit to see that

$$\begin{aligned}\mu^*(E) &\geq \sum_{i=1}^\infty \mu^*(E\cap A_i) + \mu^*\left(E\cap\left(\bigcup_{i=1}^\infty A_i\right)^c\right)\\ &\geq \mu^*\left(E\cap\left(\bigcup_{i=1}^\infty A_i\right)\right) + \mu^*\left(E\cap\left(\bigcup_{i=1}^\infty A_i\right)^c\right)\end{aligned}$$

We see that $\bigcup_{i=1}^\infty A_i \in \mathcal{A}$. Thus, $\mathcal{A}$ is a σ-algebra. Then, using the inequality above, as we see

$$\mu^*(E) \geq \sum_{i=1}^\infty \mu^*(E\cap A_i) + \mu^*\left(E\cap\left(\bigcup_{i=1}^\infty A_i\right)\right)$$

for all E, letting $E = \bigcup_{i=1}^\infty A_i$, we see that

$$\mu^*\left(\bigcup_{i=1}^\infty A_i\right) \geq \sum_{i=1}^\infty \mu^*(A_i)$$

so along with subadditivity, this implies that μ^* when restricted to $\mathcal{A}$ is countably additive, meaning that μ^* is a measure on $\mathcal{A}$.

All that is left to show is that $\mathcal{A}$ contains all of the null sets of μ^* (which are the sets for which $\mu^*(A) = 0$). Now, if $\mu^*(A) = 0$, then we immediately see

$$\mu^*(E) \geq \mu^*(E\cap A^c) = \mu^*(E\cap A) + \mu^*(E\cap A^c)$$

so thus, A is μ^*-measurable!

■

Now, we can define the Lebesgue measure formally using the outer-measure we constructed earlier!

Definition 5.2: The Lebesgue Measure

The *Lebesgue measure* on $\mathbb{R}$ is the measure given by restricting the outer measure

$$\lambda^*(A) = \inf\{\sum_{i=1}^\infty |I_i| \,:\, A\subseteq \cup I_i\}$$

to the collection of λ^*-measurable sets. The corresponding σ-algebra $\mathcal{L}$ is called the *Lebesgue σ-algebra*.

Now, we just went through all that work to define a σ-algebra for an outer measure, but why didn't we just show that the outer measure was a measure on all sets? Well, it turns out we can't always do that, as we will soon show. First, a lemma that will help our construction.

Lemma 5.3

If $A \subseteq \mathbb{R}$ and $x \in \mathbb{R}$, then if $A + x = \{a + x : a \in A\}$, then $\lambda^*(A + x) = \lambda^*(A)$. Thus, the Lebesgue measure is translation invariant.

Proof: This is quite obvious from the definition of λ^*.

Theorem 5.4: Existence of Non-Measurable Sets

There exists sets in $\mathcal{P}(\mathbb{R})$ that are not Lebesgue measurable.

Proof: First, define an equivalence relation $\sim$ on $[0, 1]$ by $x \sim y$ if, and only if, $x - y \in \mathbb{Q}$. Now, define V by choosing one representative from each equivalence class of $\sim$.

Now, suppose that V is Lebesgue-measurable. Enumerate $\mathbb{Q} \cap [-1, 1]$ as $q_1, q_2, \cdots$ and let $V_n = V + q_n$. It can be easily shown that all the V_n are disjoint, and by our lemma. $\lambda(V_n) = \lambda(V)$. Thus, we see that

$$\lambda\left(\bigcup_{n=1}^{\infty} V_n\right) = \sum_{n=1}^{\infty} \lambda(V_n) = \sum_{n=1}^{\infty} \lambda(V)$$

Thus, this must be either 0 or infinity depending on whether $\lambda(V) = 0$ or $\lambda(V) > 0$. We will show that neither can happen.

First, since every $x \in [0, 1]$ must lie in some V_n, we see that $[0, 1] \subseteq \bigcup_{n=1}^{\infty} V_n$, we must have that $\lambda(\cup_{i=1}^{\infty} V_n) \geq 1$, so $\lambda(V) = 0$ cannot occur. Then, as $V_n \subseteq [-1, 2]$ for all n, we must also have that $\lambda(\bigcup_{n=1}^{\infty} V_n) \leq \lambda([-1, 2]) \leq 3$, so $\lambda(V) > 0$ cannot occur as well. Thus, we have a contradiction, so V must not be measurable.

■

This doesn't look good... What is the point of a function that could possibly fail to give a measure to a set that we want to measure? Well, fortunately, most of the sets we care about are measurable, as we will see next.

Proposition 5.5

Every Borel set is Lebesgue measurable.

Proof: It suffices to show that each interval is Lebesgue measurable. We want to show that, for all E,

$$\lambda^*(E) \geq \lambda^*(E \cap (a, b)) + \lambda^*(E \cap (a, b)^c)$$

Let $\epsilon > 0$. By definition of λ^*, there exists a collection $\{I_n\}$ such that $\sum_{n=1}^{\infty} |I_n| \leq \lambda^*(E) + \epsilon$ and $E \subseteq \bigcup_{n=1}^{\infty} I_n$. Then, as we have

$$\lambda^*(E \cap (a, b)) \leq \sum_{i=1}^{\infty} \lambda^*(I_n \cap (a, b)) \leq \sum_{i=1}^{\infty} |I_n \cap (a, b)|$$

and

$$\lambda^*(E \cap (a, b)^c) \leq \sum_{i=1}^{\infty} |I_n \cap [b, \infty)| + \sum_{i=1}^{\infty} |I_n \cap (-\infty, a]|$$

we see that

$$\begin{aligned}\lambda^*(E \cap (a,b)) + \lambda^*(E \cap (a,b)^c) &\leq \sum_{i=1}^{\infty} (|I_n \cap (a,b)| + |I_n \cap (-\infty, a]| + |I_n \cap [b,\infty)|) \\ &= \sum_{i=1}^{\infty} |I_n| \\ &\leq \lambda^*(E) + \epsilon\end{aligned}$$

as this holds for all $\epsilon > 0$, we have

$$\lambda^*(E) \geq \lambda^*(E \cap (a,b)) + \lambda^*(E \cap (a,b)^c)$$

which was to be proved.

■

Proposition 5.6

There exists Lebesgue measurable sets that are not Borel. That is

$$\mathcal{B} \subsetneq \mathcal{L} \subsetneq \mathcal{P}(\mathbb{R})$$

Proof Sketch: We won't prove this in it's entirety, but it turns out that $|\mathcal{B}| = |\mathbb{R}|$, since we can map sequences of $\mathbb{N}$ into $\mathcal{B}$. Then, it turns out that $\mathcal{L} > \mathcal{R}$, so there are quite a bit more Lebesgue measurable sets than Borel measurable sets.

■

9/9/26 Lecture 6: Properties of the Lebesgue Measure

Proposition 6.1

There exists a set in $\mathcal{L}$ that is not in $\mathcal{B}$.

Proof: First, we assume the fact that $|\mathcal{B}| = |\mathbb{R}|$.

Now, let C be the middle thirds Cantor set. We know that $|C| = |\mathbb{R}|$. Now, every subset of C is Lebesgue measureable, since C is a null set of λ^*, so $\mathcal{P}(C) \subseteq \mathcal{L}$. However, by Cantor's theorem, we see that $|\mathcal{P}(C)| > |C| = |\mathbb{R}|$, so

$$|\mathcal{L}| \geq |\mathcal{P}(C)| > |C| = |\mathbb{R}| = |\mathcal{B}|$$

so as $|\mathcal{B}| < |\mathcal{L}|$, the desired result holds.

■

It turns out that this construction entirely highlights the issue with $\mathcal{B}$: it doesn't contain all the null sets of λ, which we will soon see.

Definition 6.2: Complete Metric Spaces

A measure space $(X, \mathcal{A}, \mu)$ is *complete* if $B \in \mathcal{A}$ and $\mu(B) = 0$ implies that for all $A \subseteq B$, $A \in \mathcal{A}$.

Notice that we can define a measure space $(\mathbb{R}, \mathcal{B}, \lambda)$ to be a measure space by restricting the Lebesgue measure to $\mathcal{B}$, but this is not a complete measure space.

Proposition 6.3

Let $A \subseteq [0,1]$ be λ-measurable. Then,

1. For any $\epsilon > 0$, there exists an open set $\mathcal{U}$ such that $A \subseteq \mathcal{U}$ and $\lambda(\mathcal{U} \backslash A) < \epsilon$.
2. For any $\epsilon > 0$, there exists a closed set $\mathcal{C}$ such that $\mathcal{C} \subseteq A$ and $\lambda A \backslash \mathcal{C} < \epsilon$.
3. There are open sets
$$\mathcal{U}_1 \supseteq \mathcal{U}_2 \supseteq \mathcal{U}_3 \supseteq \cdots \supseteq A$$
so that $\lambda\left(\bigcap_{i=1}^{\infty} \mathcal{U}_i \backslash A\right) = 0$.
4. There are closed sets
$$\mathcal{C}_1 \subseteq \mathcal{C}_2 \subseteq \cdots \subseteq A$$
such that $\lambda\left(A \backslash \bigcup_{i=1}^{\infty} E_n\right) = 0$.

Proof:
1) From the definition of λ, one can find an open set $\mathcal{U}$ such that $A \subseteq \mathcal{U}$ and $\lambda(U) < \lambda(A) + \epsilon$. Then, as $\lambda(\mathcal{U} \backslash A) = \lambda(\mathcal{U}) - \lambda(A)$ as $A \subseteq \mathcal{U}$, we see that $\lambda(\mathcal{U} \backslash A) < \epsilon$.

2) Now, consider the set $[0,1] \backslash A$, and using 1 find an open set $\mathcal{O} \supseteq [0,1] \backslash A$ such that $\lambda(\mathcal{O} \backslash ([0,1] \backslash A)) < \epsilon$.

Then, as

$$\begin{aligned}\lambda(\mathcal{O}\backslash([0,1]\backslash A)) &= \lambda(\mathcal{O}) - (1-\lambda(A)) \\ &= \lambda(A) - (1-\lambda(\mathcal{O})) \\ &= \lambda(A) - \lambda([0,1]\backslash\mathcal{O})\end{aligned}$$

and we see that $[0,1]\backslash\mathcal{O}$ is a closed set contained in A, we see that this last expression is equal to $\lambda(A\backslash([0,1]\backslash\mathcal{O})) < \epsilon$.

3) By 1, there exists open sets $V_n \supseteq A$ such that $\lambda(V_n\backslash A) < 1/n$. Let $U_n = \bigcap_{k=1}^{n} V_n$. Then, we see that

$$U_1 \supseteq U_2 \supseteq \cdots \supseteq A$$

the U_n are open, and $\lambda\left(\bigcap_{n=1}^{\infty} V_n\backslash A\right) = \lim_{n\to\infty} \lambda(U_n\backslash A)$, which equals zero as $\lambda(U_n\backslash A) < 1/n$ for each n (as $U_n \subseteq V_n$).

4) Similar proof as **3)**.

■

Corollary 6.4

$\mathcal{L}$ is generated by $\mathcal{B}$ and the null sets of λ.

Proof: Let $A \in \mathcal{L}$. From the proposition above, there exists a $G_n = \cap_{i=1}^{\infty}\mathcal{U}_i \in B$ so that $G_n \supseteq A_n = A \cap [n, n+1]$ and $\lambda(G_n\backslash A_n) = 0$. Thus, we see that $\bigcup_{i=1}^{\infty} G_n \in \mathcal{B}$ and

$$A = \left(\bigcup_{i=1}^{\infty} G_n\right) \setminus \left(\bigcup_{i=1}^{P} \infty G_n\backslash A_n\right)$$

so A can be written as the union, complement, and intersection of Borel and null sets.

■

Definition 6.5: G_δ and F_σ Sets

A countable intersection of open sets is a G_δ set. A countable union of closed sets is a F_σ set.

The Lebesgue-Stieljes Measure

Let $\mathcal{C}$ be a collection of intervals of the form $(a,b]$ and let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous. Then, define $\ell((a,b]) = \alpha(b) - \alpha(a)$. Now, for $E \subseteq \mathbb{R}$, define

$$m^*(E) = \inf\{\sum_{n=1}^{\infty} \ell(A_n) \,:\, A_n \in \mathcal{C} \text{ and } E \subseteq \bigcup_{n=1}^{\infty} A_n\}$$

For example, if $\alpha = x$, then we get the Lebesgue outer measure, and if α is zero for all $x < 0$ and 1 for all $x \geq 0$, we see that m^* is the point mass outer measure.

Proposition 6.6

The function $m^* : \mathcal{P} \to [0, \infty]$ is an outer measure. Let m be the measure defined by the restriction of m^* to m^*-measurable sets. Then, every Borel set is m^*-measurable and $m^*((a, b]) = \ell(a, b]$.

Proof: This is the same proof for λ^*, but we replace $|(a, b)| = b - a$ with $\alpha(b) - \alpha(a)$.

9/11/26 Lecture 7: L-S Measures and More

Remark: Before we prove that L-S measures satisfy the conclusions of Proposition 6.3, recall that given a L-S outer measure m^*, all Borel sets are m^*-measurable, but the σ-algebra of m^*-measurable sets is not necessarily the Lebesgue σ-algebra.

Proposition 7.1

Let m be an L-S measure. Then, if $A \subseteq [0,1]$, then for all $\epsilon > 0$, there exists an open set $\mathcal{U} \supseteq A$ such that $m(\mathcal{U} \backslash A) < \epsilon$. With this, the rest of the conclusions of Proposition 6.3 also follow.

Proof: By definition, we can write $A \subseteq \bigcup_n (a_n, b_n]$ such that $\sum_{i=1}^{\infty} \ell(a_n, b_n] \leq m(A) + \frac{\epsilon}{2}$. Now, for each n, notice that

$$m(a_n, b_n] = \lim_{m \to 0} m(a_n, b_n + \frac{1}{m})$$

as we see that $(a_n, b_n + \frac{1}{m}) \downarrow (a_n, b_n]$. Thus, we can find a c_n such that $m(a_n, c_n) \leq m(a_n, b_n] + \frac{\epsilon}{2^{n+1}}$. Thus, we have that

$$\begin{aligned}
m(\bigcup_{n=1}^{\infty} (a_n, c_n) \backslash A) &= m(\bigcup_{n=1}^{\infty} (a_n, c_n)) - m(A) \\
&\leq \sum_{i=1}^{\infty} m((a_n, c_n)) - m(A) \\
&\leq \sum_{i=1}^{\infty} \left(m(a_n, b_n] + \frac{\epsilon}{2^{n+1}} \right) - m(A) \\
&= \sum_{i=1}^{\infty} \ell(a_n, b_n] + \frac{\epsilon}{2} - m(A) \\
&\leq m(A) + \frac{\epsilon}{2} + \frac{\epsilon}{2} - m(A) \\
&= \epsilon
\end{aligned}$$

which was to be shown as $\bigcup_{n=1}^{\infty} (a_n, c_n)$ is open.

■

Now, in these past few lectures we've developed how to go from an outer-measure to a measure on a σ-algebra, but what if we know a bit more about our "measuring function". What if it's already pretty close to a measure? We will explore that now.

Definition 7.2: Algebras

Given a set X, a *algebra* $\mathcal{A}_0$ is a collection of subsets such that

1. $\emptyset, X \in \mathcal{A}_0$
2. $A^c \in \mathcal{A}_0$ if $A \in \mathcal{A}_0$
3. If $A_1, \ldots, A_n \in \mathcal{A}_0$, then $\bigcup_{i=1}^{n} A_i \in \mathcal{A}_,$

Definition 7.3: Premeasures

A *premeasure* ℓ on an algebra $\mathcal{A}_0$ is a map $\ell : \mathcal{A}_0 \to [0, \infty]$ such that

1. $\ell(\emptyset) = 0$
2. If $A_1, \ldots \in \mathcal{A}_0$ and $\bigcup_{i=1}^{\infty} A_n \in \mathcal{A}_0$, and the A_i are pairwise disjoint, then

$$\ell\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \ell(A_n)$$

Now, one last definition that we will need for our extension theorem.

Definition 7.4: Finite and σ-Finite Measures

A measure μ on X is *finite* if $\mu(X) < \infty$. A measure μ on X is *σ-finite* if there exists $X_n \in \mathcal{A}$ such that $X = \bigcup_{n=1}^{\infty} X_n$ and $\mu(X_n) < \infty$.

Thus, for example λ on $\mathbb{R}$ is not finite but it is σ-finite. Most spaces (and measures) that we will be concerned with are σ-finite. Now, we move onto our extension theorem.

Theorem 7.5: Caratheodory Extension Theorem

Let $\mathcal{A}_0$ be an algebra on X and let ℓ be a premeasure on $\mathcal{A}_0$. Define

$$\mu^*(E) = \inf\left\{\sum_{i=1}^{\infty} \ell(A_n) \,:\, A_n \in \mathcal{A}_\prime, E \subseteq \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}_0\right\}$$

Then,

1. μ^* is an outer measure.
2. $\mu^*(A) = \ell(A)$ if $A \in \mathcal{A}_0$.
3. Every μ^* null set is μ^*-measurable and every $A \in \mathcal{A}_0$ is μ^*-measurable.
4. If ℓ is σ-finite, then μ^* is the unique extension of ℓ to $\sigma(\mathcal{A}_0)$.

Proof: In this lecture we will prove the first three statements, and we will prove the fourth one in the next lecture.

1) This is the same proof as for our other extension theorems.

2) Suppose that $E \in \mathcal{A}_0$ and $E \subseteq \bigcup_{i=1}^{\infty} A_n$ for $A_n \in \mathcal{A}_0$. Define $B_n = E \cap \left(A_n \setminus \bigcup_{i=1}^{n-1} A_i\right)$. We see that $B_n \in \mathcal{A}$ for each n.

Then, we see that $\ell(E) = \sum_n \ell(B_n) \leq \sum_n \ell(A_n)$, so taking infimums over such collections, we see that $\ell(E) \leq \mu^*(E)$, and as we clearly have that $\mu^*(E) \leq \ell(E)$, we have the desired property.

3) Let $A \in \mathcal{A}_0$ and let $E \subseteq X$. WE want to show that $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Choose $B_n \in \mathcal{A}_0$

so that $\bigcup_n B_n \in \mathcal{A}_\prime$, $E \subseteq \bigcup_n B_n$, and

$$\sum_n \ell(B_n) \leq \mu^*(E) + \epsilon$$

Then, we see that

$$\begin{aligned}
\mu^*(E) + \epsilon &\geq \sum_n \ell(B_n \cap A) + \ell(B_n \cap A^c) \\
&= \sum_n \ell(B_n \cap A) + \sum_n \ell(B_n \cap A^c) \\
&\geq \mu^*(E \cap A) + \mu^*(E \cap A^c)
\end{aligned}$$

where the nontrivial step of the proof is that $\ell(B_n) = \ell(B_n \cap A) + \ell(B_n \cap A^c)$ as $A \in \mathcal{A}_0$. Thus, as this holds for all ϵ, we have what was to be shown.

■

9/14/26 Lecture 8: Intro to the Lebesgue Integral

First, we finish up the proof for Caratheodory's theorem seen above.

Proof of 4) Let ν be a measure on $\sigma(\mathcal{A}_0)$ such that $\nu(A) = \ell(A)$ for all $A \in \mathcal{A}_\prime$. WE want to show that $\nu(A) = \mu^*(A)$ for all $A \in \sigma(A_n)$.

First, we will consider the case when μ^* is finite. Fix $A \in \sigma(\mathcal{A}_0)$ and let $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}_\prime$. Then, as

$$\nu(A) \leq \sum_{n=1}^{\infty} \nu(A_n) = \sum_{n=1}^{\infty} \ell(A)$$

taking infs over such covers, we see that $\nu(A) \leq \mu^*(A)$.

Now, we want to show that ν and μ^* agree on sets of the form $\bigcup_n A_n$ for $A_n \in \mathcal{A}_0$. Let

$$B_n = \bigcup_{j=1}^{n} A_j$$

then, we see that

$$\mu^*(\cup_{i=1}^{\infty} A_i) = \lim_{n\to\infty} \mu^*(B_n) = \lim_{n\to\infty} \nu(B_n) = \nu(\cup_{i=1}^{\infty} A_i)$$

Now, choose A_n so that $A \subseteq \bigcup_{i=1}^{\infty} A_n$ and $A_n \in \mathcal{A}_0$ such that $\mu^*(A) + \epsilon \geq \sum_n \mu^*(A_n)$. Then,

$$\begin{aligned} \mu^*(A) + \epsilon &\geq \sum_n \mu^*(A_n) \\ &\geq \mu^*\left(\bigcup_{i=1}^{\infty} A_n\right) \end{aligned}$$

thus, we see that $\epsilon \geq \mu^*\left(\bigcup_{i=1}^{\infty} A_n\right) - \mu^*(A) = \mu^*(\bigcup_{i=1}^{\infty} A_n \backslash A)$ (this step requires that μ^* is finite). Then, we see that

$$\begin{aligned} \mu^*(A) \leq \mu^*(\bigcup_{i=1}^{\infty} A_n) = \nu(\bigcup_{i=1}^{\infty} A_n) \\ &= \nu(A) + \nu(\bigcup_{i=1}^{\infty} A_n \backslash A) \\ &\leq \nu(A) + \mu^*(\bigcup_{i=1}^{\infty} A_n \backslash A) \\ &< \nu(A) + \epsilon \end{aligned}$$

where the second to last step came from above. Thus, $\mu^*(A) \leq \nu(A) + \epsilon$ for all ϵ, so $\mu^*(A) \leq \nu(A)$.

Now, if μ^* is σ-finite, then we can write $X = \bigcup_{n=1}^{\infty} X_n$ with $X_n \uparrow X$ with $\mu^*(X_n) < \infty$. Then, if we consider the measures $\mu^*|_{X_n}$ and $\nu|_{X_n}$ (where $\mu|_{X_n}(A) = \mu(X_n \cap A)$), we see that as $\mu^*|_{X_n}$ and $\nu|_{X_n}$ are finite measures, they have to agree by the argument above. Then, for $A \in \sigma(\mathcal{A}_\prime)$, we see that

$$\mu^*(A) = \lim_{n\to\infty} \mu^*(A \cap X_n) = \lim_{n\to\infty} \nu(A \cap X_n) = \nu(A)$$

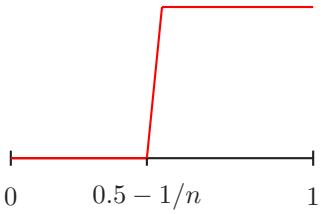
which was desired.

■

Now, lets move on to functions. As a motivation, consider $C[0,1]$ with the metric

$$d(f,g) = \int_0^1 |f(x) - g(x)|\, dx$$

This is nice until you realize that it doesn't make $C[0,1]$ complete! Consider the functions as described below



These turn out to be Cauchy under this metric, but they converge to the function

$$f(x) = \chi_{[1/2,1]}$$

where χ_A is the characteristic function of A which is 1 on A and 0 off of A. The problem is that $\chi_{[1/2,1]}$ is not continuous, so it cannot be the limit. This example is somewhat tame in that the limiting function is still Riemann integrable, but there are some Cauchy sequences in $C[0,1]$ that converge to non Riemann integrable functions, so we want to see how to fix that.

First, consider $\chi_{\mathbb{Q}\cap[0,1]}$. This is not Riemann integrable as $U(\chi_{\mathbb{Q}\cap[0,1]}, P) = 1$ and $L(\chi_{\mathbb{Q}\cap[0,1]}, P) = 0$ for any partition. However, maybe we should still try to make sense of it's integral.

Question: What should $\int \chi_{\mathbb{Q}\cap[0,1]} dx$ be?
Intuitive Answer Since $\int$ is the area under a function, $\int \chi_E$ should be the measure of E for any set E. That is, $\int \chi_E = \lambda(E)$. Then, as $\lambda(\mathbb{Q}\cap[0,1]) = 0$, it would be natural to expect that $\int \chi_{\mathbb{Q}\cap[0,1]} dx = 0$. How do we formalize this? First, we start by defining the integral on characteristic functions as we did above.

Definition 8.1: Lebesgue Integral of a Characteristic Function

Given a measure space $(X, \mathcal{A}, \mu)$, define the *Lebesgue integral* of a characteristic function χ_A for $A \in \mathcal{A}$ to be

$$\int \chi_A d\mu = \mu(A)$$

Definition 8.2: Simple Functions

A *simple function* is a function of the form

$$f = \sum_{i=1}^{\infty} c_n \chi_{A_n}$$

Then, as we want our integral to be linear as well, we can define

$$\int \sum_{n=1}^{\infty} a_n \chi_{A_n} d\mu = \sum_{n=1}^{\infty} a_n \mu(A_n)$$

for $A_n \in \mathcal{A}$.

Now, heres a definition that seems random right now, but we will soon see it describes exactly what we want.

Definition 8.3: Measurable Functions

A function $f : X \to \mathbb{R}$ is *measurable* if $\{x \in X \,:\, f(x) > a\} = f^{-1}(a, \infty) \in \mathcal{A}$ for all $a \in \mathbb{R}$.

Now, we see, considering some χ_A, that $f^{-1}(a, \infty) = \emptyset$ if $a \geq 1$, $f^{-1}(a, \infty) = A$ if $0 \leq a < 1$, and $f^{-1}a, \infty = X$ if $a < 0$. Thus, we see that χ_A is measurable if A is a measurable set!

9/18/26 Lecture 9: Measurable Functions

Proposition 9.1

The following are equivalent.

1. f is measurable.
2. $f^{-1}((-\infty, a]) \in \mathcal{A}$ for all a.
3. $f^{-1}((-\infty, a)) \in \mathcal{A}$ for all a.
4. $f^{-1}([a, \infty)) \in \mathcal{A}$ for all a.

Proof: The idea is that f^{-1} preserves complements, unions, and intersections.

1) $\to$ 2) Since $f^{-1}((a, \infty))^c = f^{-1}((-\infty, a])$, we see that $f^{-1}((-\infty, a]) \in \mathcal{A}$.

2) $\to$ 3) Writing $(-\infty, a) = \bigcup_{n=1}^{\infty} (-\infty, a - 1/n]$, we see that

$$f^{-1}((-\infty, a)) = \bigcup_{n=1}^{\infty} f^{-1}((-\infty, a - 1/n]) \in \mathcal{A}$$

3) $\to$ 4) Since $f^{-1}((-\infty, a))^c = f^{-1}([a, -\infty))$, we see that $f^{-1}([a, \infty)) \in \mathcal{A}$.

4) $\to$ 1) Writing $(a, \infty) = \bigcup_{n=1}^{\infty} [a + 1/n, \infty)$, we see that

$$f^{-1}((a, \infty)) = \bigcup_{n=1}^{\infty} f^{-1}([a + 1/n, \infty)) \in \mathcal{A}$$

■

Proposition 9.2

Suppose that X is a metric space and let $\mathcal{A}$ be a σ-algebra on X that contains every open set. Then, every continuous function $f : X \to \mathbb{R}$ is measurable.

Proof: Since (a, ∞) is open, $f^{-1}((a, \infty))$ is open, so it lies in $\mathcal{A}$.

■

Proposition 9.3

Suppose that $f, g : X \to \mathbb{R}$ are measurable. Then, so are the following:

1. $f + g$
2. cf for $c \in \mathbb{R}$
3. fg
4. $\max(f, g)$
5. $\min(f, g)$

Proof:

1) We will show that

$$(f+g)^{-1}(-\infty, a) = \bigcup_{r \in \mathbb{Q}} (f^{-1}(-\infty, a-r) \cap g^{-1}(-\infty, r])$$

which is sufficient as the LHS lies in $\mathcal{A}$. First, it is quite clear that we have the backwards inclusion. Now, let $x \in (f+g)^{-1}(-\infty, a)$. Then $f(x) + g(x) < a$, so $f(x) < a - g(x)$. Then, by density of $\mathbb{Q}$, we see there exists an $r \in \mathbb{Q}$ for which $f(x) < a - r \leq a - g(x)$, from which we can also include that $g(x) \leq r$. Thus, we see that $x \in f^{-1}(-\infty, a-r) \cap g^{-1}(-\infty, r]$, so we have the forward inclusion.

2) If $c = 0$, this is trivial, so suppose not. Then, we see that

$$(cf)^{-1}(-\infty, a) = \{x : cf(x) < a\} = \begin{cases} \{x : f(x) < \frac{a}{c}\} & c > 0 \\ \{x : f(x) > \frac{a}{c}\} & c < 0 \end{cases}$$

which are measurable sets in all cases.

3) First, notice that f^2 is measurable since, for $a < 0$, $(f^2)^{-1}(a, \infty) = X$, and for $a \geq 0$,

$$(f^2)^{-1}(a, \infty) = \{x : f(x) > \sqrt{a}\} \cup \{x : f(x) < -\sqrt{a}\}$$

Then, fg is measurable as we can write

$$fg = \frac{1}{2}\left[(f+g)^2 - f^2 - g^2\right]$$

4) We see that

$$(\max(f, g))^{-1}(a, \infty) = f^{-1}(a, \infty) \cup g^{-1}(a, \infty) \in \mathcal{A}$$

5) We see that

$$(\min(f, g))^{-1}(-\infty, a) = f^{-1}(-\infty, a) \cup g^{-1}(-\infty, a) \in \mathcal{A}$$

■

Proposition 9.4

If f_n is a sequence of measurable functions, then $\sup_n f_n, \inf_n f_n, \limsup_n f_n$ and $\liminf_n f_n$ are measurable.

Proof: We can write

$$(\sup_n f_n)^{-1}(a,\infty) = \bigcup_{n\geq 1} f_n^{-1}(a,\infty) \in \mathcal{A}$$

Then, as $\inf_n f_n = -\sup_n(-f_n)$, we see that $\inf_n f_n$ is also measurable. Then, as

$$\limsup_n f_n = \inf_n \sup_{k\geq n} f_k \quad \text{and} \quad \liminf_n f_n = \sup_n \inf_{k\geq n} f_k$$

we see that $\limsup_n f_n$ and $\liminf_n f_n$ are measurable.

Also, note that if $\lim_n f_n$ exists, then it is equal to $\limsup_n f_n$, so it is also measurable.

Definition 9.5: Equality Almost Everywhere

We say that two measurable functions f and g are *equal almost everywhere*, denoted $f \overset{\text{a.e.}}{=} g$, if $\{x : f(x) \neq g(x)\}$ has measure 0.

Note that the above set is measurable as it is equal to $\{x : (f-g)(x) \geq 0\} \cap \{x : (f-g)(x) \leq 0\}$. An example of the above definition is that $\chi_{\mathbb{Q}}$ is equal to 0 almost everywhere because $\mathbb{Q}$ is the set on which they differ.

Definition 9.6: Convergence Almost Everywhere

If f_n is a sequence of measurable functions, we say f_n *converges to* f *almost everywhere*, denoted $f_n \to f$ a.e., if

$$\{x : \lim_{n\to\infty} f_n(x) \neq f(x)\}$$

has measure zero.

Definition 9.7: Borel Measurability

Let X be a metric space and $\mathcal{B}(X)$ be it's Borel σ-algebra. Then, $f : X \to \mathbb{R}$ is *Borel measurable* if it is measurable with respect to $\mathcal{B}(X)$. That is, $f^{-1}(a,\infty) \in \mathcal{B}(X)$ for all a.

9/16/26 Lecture 10: More on Lebesgue Integrals

Proposition 10.1

If $f : \mathbb{R} \to \mathbb{R}$ is monotone, then it is Borel-measurable.

Proof: WLOG, assume that f is increasing. We want to show that $f^{-1}(a, \infty)$ is an interval. Now, let

$$x_0 = \sup\{x \,:\, f(x) \leq a\}$$

We have to cases. If $f(x_0) \leq a$, then $f^{-1}(a, \infty) = (x_0, \infty)$. Similarly, if $f(x_0) > a$, then $f^{-1}(a, \infty) = [x_0, \infty]$.

Proposition 10.2

Let $(X, \mathcal{A})$ be a measurable space. If $f : X \to \mathbb{R}$ is measurable, then $f^{-1}(A) \in \mathcal{A}$ whenever $A \in \mathcal{B}$.

Proof: Let $\mathcal{C} = \{A \subseteq \mathbb{R} \,:\, f^{-1}(A) \in \mathcal{A}\}$. We see that $\mathcal{C}$ contains all subsets of the form (a, ∞). Now, we see that $\mathbb{R}$ and $\emptyset \in \mathcal{C}$, it is closed under complements, and is closed under unions (as f^{-1} distributes across them). Thus, we see that $\sigma(\{(a, \infty)\}) \subseteq \mathcal{C}$, so $\mathcal{B} \subseteq \mathcal{C}$.

■

Proposition 10.3

Let $f \geq 0$ and measurable. Then, there exists a sequence of simple functions $s_n \geq 0$ such that $s_n \uparrow f$. That is, $0 \leq s_1 \leq s_2 \leq \cdots \leq f(x)$ and $\sup_n s_n = f$.

Proof: For each n, define

$$I_{j,n} = \left[\frac{j-1}{2^n}, \frac{j}{2^n}\right)$$

now for $j = 1, \ldots, n2^n$, let

$$s_n = \sum_{j=1}^{n2^n} \frac{j-1}{2^n} \chi_{f^{-1}(I_{j,n})} + n\chi_{f^{-1}[n,\infty)}$$

Now, it can be checked that s_n is increasing, and that for each x, $s_n(x) \uparrow f(x)$.

Now, we will work towards proving Lusin's theorem.

Definition 10.4: Support of a Function

The *support* of a function f is

$$\operatorname{supp}(f) = \overline{\{x \,:\, f(x) \neq 0\}}$$

Now, one way to define continuity is that, for $f : X \to \mathbb{R}$, f is continuous if $f(x_n) \to f(x)$ for a sequence $x_n \to x$.

If $Y \subseteq X$, then $f|_Y : Y \to \mathbb{R}$ is given by $f|_Y(x) = f(x)$. If we have a function f defined only on Y, then continuity means that $g(x_n) \to g(x)$ but only for $x_n \to x$ contained in Y.

Example 1

Consider $\chi_{\mathbb{Q}} : \mathbb{R} \to \mathbb{R}$. This is not continuous, but we see that $\chi_{\mathbb{Q}}|_{\mathbb{Q}} : \mathbb{Q} \to \mathbb{R}$ is continuous as it is the constant function 1.

Lemma 10.5

If $A \subseteq \mathbb{R}$, define the distance from a point to a set as

$$d(x, A) := \inf\{d(x, y) : y \in A\}$$

This is a continuous function, and if A is closed, then $f(x) = 0$ if, and only if, $x \in A$.

Now, we state and prove Lusin's Theorem.

Theorem 10.6: Lusin's Theorem

Let $f : [0, 1] \to \mathbb{R}$ be Lebesgue measurable. Then, for any $\epsilon > 0$, there exists a closed set $E \subseteq [0, 1]$ such that $\lambda([0, 1]\backslash E) < \epsilon$ and $f|_E$ is continuous.

Proof: First, we start by proving it for characteristic functions. Let $f = \chi_A$. Then, we can find $K \subseteq A \subseteq U$ with K closed and U open so that $\lambda(A\backslash K) < \epsilon/2$ and $\lambda(U\backslash A) < \epsilon/2$.

Let $E = K \cup (U^c) \cap [0, 1]$. We see that E is closed, so we need to show that $f|_E$ is continuous. Let

$$g(x) = \frac{d(x, U^c)}{d(x, K) + d(x, U^c)}$$

This function is continuous because it's components are continuous and $d(x, K) + d(x, U^c) = 0 \implies d(x, K) = 0$ and $d(x, U^c) = 0$, which isn't possible as $K \cap U^c = \emptyset$ and they are both closed.

Then, notice that

$$g|_E(x) = \begin{cases} 0 & x \in U^c \\ 1 & x \in K \end{cases} = f(x)$$

so as $g|_E$ is continuous, we see that f is continuous. Also, notice that

$$\lambda([0, 1]\backslash E) = \lambda(K^c \cap U \cap [0, 1]) \leq \lambda(K^c \cap U) = \lambda(U\backslash K) < \epsilon$$

Now, we will prove it for simple functions. Let

$$f = \sum_{n=1}^{N} a_n \chi_{A_n}$$

Now, by the previous step, there exist for each n, a K_n so that $\lambda([0, 1]\backslash K_n) < \frac{\epsilon}{n}$, and $\chi_{A_n}|_{K_n}$ is continuous. Then, let

$$E = \bigcap_{n=1}^{N} K_n$$

We see that E is closed, $\mu([0, 1]\backslash E) < \epsilon$, and as each χ_{A_n} is continuous on E, we see that $f|_E$ is continuous.

We will finish this proof next time.-

9/21/26 Lecture 11: More on Integrals

We first start by finishing the proof of Lusin's theorem.

Proof of Lusin's Theorem:

Now, we will prove this for positive functions f bounded above by N. Let

$$\tilde{s}_n = \sum_{j=1}^{N2^N+1} \frac{j-1}{2^n} \chi_{f^{-1}[\frac{j-1}{2^n}, \frac{j}{2^n})}$$

This is a similar construction as before, but as N is bounded, we don't need the extra term at the end. Let $f_n = \tilde{s}_{n+1} - \tilde{s}_n$. This is still a simple function, and satisfies

$$|f_n(x)| < \frac{1}{2^n}$$

Then, as f_n are simple functions, we can find sets E_n with $\lambda([0,1]\backslash E_n) < \frac{\epsilon}{2^{n+1}}$. Now, we see that $f_n|_{E_n}$ is continuous, and notice that

$$\tilde{s}_0(x) + \sum_{j=1}^{n} f_j(x) = \tilde{s}_{n+1}(x)$$

Now, letting $E = \bigcap_{n=1}^{\infty} E_n$, then we see that each f_n is continuous on E, and we see that $(\sum_{j=1}^{\infty} f_n(x) - \tilde{s}_0(x))$ converges uniformly to f by the Weierstrass M-test. Thus, f is continuous on E, as each f_n is continuous on E.

Since we see that

$$\lambda\left([0,1]\backslash \bigcap_{i=1}^{n} E_n\right) \leq \sum_{i=1}^{\infty} \lambda([0,1]\backslash E_n) < \epsilon$$

we are done.

Now, suppose f is positive, but not necessarily bounded. Let

$$B_N(x) = \{x : f(x) \leq N\}$$

Notice that $[0,1] = \bigcup_{n=1}^{\infty} B_N$. Now, we can find an N for which

$$\lambda([0,1]\backslash B_n) < \frac{\epsilon}{3}$$

and further find a closed $K \subseteq B_N$ so that

$$\lambda(B_N \backslash K) < \epsilon/3$$

Consider $f\chi_K$. This function is positive and bounded, so we can find $E \subseteq [0,1]$ so that $f\chi_K|_E$ is continuous and $\lambda([0,1]\backslash E) < \frac{\epsilon}{3}$. Also, notice that $f|_K = f\chi_K|_K$.

Then, we see that $\lambda([0,1]\backslash(E \cap K)) < \epsilon$, and $f|_{E\cap K}$ is continuous as it agrees with $f\chi_K$ on $E \cap K$.

Finally, let f be any measurable function. Then, write $f = f_+ - f_-$ where $f_\pm = (\pm f) \vee 0$. Then, apply the previous step to f_+ and f_-, and the intersection of the sets on which they are continuous gives us a set on which f is continuous.

■

Here's something that comes directly from this theorem.

Corollary 11.1

Let $f : [0,1] \to \mathbb{R}$ be Lebesgue measurable. For every $\epsilon > 0$, there is a continuous function $g : [0,1] \to \mathbb{R}$ so that

$$\lambda(\{x \,:\, f(x) \neq g(x)\}) < \epsilon$$

Proof: By Lusin's theorem, we can find a closed set $E \subseteq [0,1]$ so that $f\,|_E$ is continuous and $\lambda([0,1]\backslash E) < \epsilon$.

Now, I want to find a g that agrees with f on E that is still continuous on its complement. Now, note that

$$[0,1]\backslash E = \bigcup_{n=1}^{\infty} (a_n, b_n)$$

now, on (a_n, b_n), we do linear interpolation. We know that $g(b_n) = f(b_n)$ and $g(a_n) = f(a_n)$. Then, on (a_n, b_n), let g be the line between $(a_n, f(a_n))$ and $(b_n, f(b_n))$. That is, let

$$g\,\big|_{(a_n,b_n)} = f(a_n) + \frac{f(b_n) - f(a_n)}{b_n - a_n}(x - a_n)$$

Now, the claim is that g is in fact continuous. We will not prove this.

■

Lebesgue Integral

Let $(X, \mathcal{A}, \mu)$ be a measure space. We will define the Lebesgue integral again, but now with more tools.

Definition 11.2: Lebesuge Integral for Simple Functions

For a simple function

$$s = \sum_{j=1}^{n} a_j \chi_{E_j},\ E_j \subseteq \mathcal{A}$$

define

$$\int s d\mu = \sum_{i=1}^{\infty} a_j \mu(E_j)$$

Note that this sum can be infinite if $\mu(E_j) = \infty$, but if $a_j = 0$ and $\mu(E_j) = \infty$, then we take by convention that $0 \cdot \infty = 0$.

Now, note that for any simple function s, there are many ways to represent it as a sum of characteristic functions. However, luckily it turns out that the above definition is independent of the representation of s.

Definition 11.3: Lebesgue Integral

Let $f \geq 0$ be measurable. Then, define

$$\int f d\mu = \sup \left\{ \int s d\mu \ : \ s \leq f, \ s \text{ is simple} \right\}$$

Now, if f is not necessarily positive, then let $f = f^+ - f^-$ and define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

but only when not both $\int f^+ d\mu$ and $\int f^- d\mu$ are infinite.

Proposition 11.4

Assume f and g are measurable. The following are true about Lebesgue integrals

1. If $0 \leq a \leq f \leq b$ for $x \in X$ and $\mu(X) < \infty$, then
$$a\mu(X) \leq \int f d\mu \leq b\mu(X)$$
2. If $0 \leq f(x) \leq g(x)$
$$0 \leq \int f d\mu \leq \int g d\mu$$
3. For any constant c,
$$\int cf d\mu = c \int f d\mu$$
4. If $\mu(A) = 0$, then
$$\int f \chi_A d\mu = 0$$

Proof:

(1) Note that the simple function $a\chi_X \leq f(x)$, and as $\int a\chi_X = a\mu(X)$, it is immediately clear that $a\mu(X) \leq \int f d\mu$ from the definition. Now, let $s \leq f$ be a simple function with $s = \sum_{j=1}^n a_j \chi_{A_j}$. We want to show that

$$\int s d\mu \leq b\mu(X)$$

Now, we want to make these A_j disjoint. Since s takes on finitely many values, we can write

$$s = \sum_{j=1}^m b_j \chi_{B_j}$$

for disjoint B_j. Now, as $b_j \leq b$ for each j, then we see that $s \leq f \leq b$ and

$$\int \sum_{j=1}^m b_j \chi_b \leq b \sum_{j=1}^m \mu(B_j) \leq b\mu(X)$$

■

9/23/26 Lecture 12: Monotone Convergence Theorem

Defintion 12.1: Integrability

We say that f is *integrable* if

$$\int |f| \, d\mu < \infty$$

Question: Given a sequence of measurable functions $\{f_n\}$ with $\lim_{n\to\infty} f_n(x) = f(x)$, when can we write

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu$$

This is not so clear. Consider the function $f_n(x) = \chi_{[n,n+1]}(x)$. We know that $\lim_{n\to\infty} f_n(x) = 0$, and $\int 0 d\mu = 0$, but for any n, we have that $\int f_n d\mu = 1$. Thus,

$$\lim_{n\to\infty} f_n d\mu \neq \int \lim_{n\to\infty} f_n$$

Here's another problem. Let $f_n(x) = n\chi_{(0,1/n)}$. We see that $\int f_n d\mu = 1$, but $f_n \to 0$.

Here's a final example. Let $f_n(x) = \frac{1}{n}\chi_{[0,n]}$. Then, $\int f_n d\mu = 1$, but $f_n \to 0$.

Theorem 12.2: Monotone Convergence Theorem

Suppose that $0 \leq f_1 \leq f_2 \leq \cdots$ is a monotone sequence of nonnegative measurable functions. Then,

$$\lim_{n\to\infty} \int f_n d\mu = \int \lim_{n\to\infty} f_n d\mu$$

Proof: Let $f(x) = \lim_{n\to\infty} f_n(x)$. Then, $f_n(x) \leq f(x)$ for all $x \in X$ and n, so we know that

$$\int f_n d\mu \leq \int f d\mu \implies \lim_{n\to\infty} \int f_n d\mu \leq \int f d\mu$$

Thus, what is left to show is that $\int f d\mu \leq \lim_{n\to\infty} \int f_n d\mu$, which is to show that if $s = \sum_{j=1}^N a_j \chi_{A_j} \leq f$, then $\int s d\mu \leq \lim_{n\to\infty} \int f_n d\mu$.

Take the A_j to be disjoint and let $0 < \epsilon < 1$. Then, for every $x \in A_j$, we see that $f_n(x) > (1-\epsilon)a_j$ for sufficiently large n. Now, let

$$A_{j,n} = \{x \in A_j \, : \, f_n(x) > (1-\epsilon)a_j\}$$

Then, notice that $A_{j,n} \uparrow A_j$. This implies that

$$\mu(A_j) = \lim_{n\to\infty} \mu(A_{j,n})$$

and as $f_n \geq \sum_{j=1}^N (1-\epsilon)a_j \chi_{A_{j,n}}$ for all n, we see that

$$\int f_n d\mu \geq \sum_{j=1}^N (1-\epsilon)a_j \mu(A_{j,n})$$

Taking limits as $n \to \infty$, we see that

$$\lim_{n\to\infty} f_n d\mu \geq \sum_{j=1}^N (1-\epsilon)a_j \mu(A_j) = (1-\epsilon)\int s d\mu$$

Thus, taking $\epsilon \to 0$, we have that

$$\lim_{n\to\infty} \int f_n d\mu \geq \int s d\mu$$

which was to be shown.

■

Corollary 12.3

If f and g are integrable, then $f+g$ is integrable and

$$\int (f+g)d\mu = \int f d\mu + \int g d\mu$$

Additionally, if f and g are non-negative but not necessarily integrable, this equality still holds.

Proof: First, suppose f and g are non-negative simple functions $f = \sum_{j=1}^n a_j \chi_{A_j}$ and $g = \sum_{j=1}^m b_j \chi_{B_j}$. Now, $f+g$ is a simple function with

$$\int (f+g)d\mu = \sum_{j=1}^n a_j \mu(A_j) + \sum_{j=1}^m b_j \mu(B_j) = \int f d\mu + \int g d\mu$$

Now, suppose f and g are non-negative. Then, we can find sequences of non-negative simple functions s_n, t_n with $s_n \uparrow f$ and $t_n \uparrow g$. Then,

$$\begin{aligned}
\int f+g d\mu &= \int \lim_{n\to\infty} s_n + t_n d\mu \\
&= \lim_{n\to\infty} \int s_n + t_n d\mu && \text{(MCT)} \\
&= \lim_{n\to\infty} \int s_n d\mu + \lim_{n\to\infty} \int t_n d\mu && \text{(need step 1!)} \\
&= \int \lim_{n\to\infty} s_n d\mu + \int \lim_{n\to\infty} t_n d\mu \\
&= \int f d\mu + \int g d\mu
\end{aligned}$$

Finally, let f and g be integrable but not necessarily nonnegative. To see that $f+g$ is integrable, notice that

$$\int |f+g|\, d\mu \leq \int |f| + |g|\, d\mu = \int |f|\, d\mu + \int |g|\, d\mu$$

so the integral must be finite. Now, write

$$(f+g)^+ - (f+g)^- = (f+g) = (f^+ - f^-) + (g^+ - g^-)$$

Thus, we see that

$$(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+$$

These are non-negative functions, so by step 2,

$$\int (f+g)^+ d\mu + \int f^- d\mu + \int g^- d\mu = \int (f+g)^- d\mu + \int f^+ d\mu + \int g^+ d\mu$$

Thus,

$$\int (f+g)^+ d\mu - \int (f+g)^- d\mu = \left(\int f^+ d\mu - \int f^- d\mu\right) + \left(\int g^+ d\mu - \int g^- d\mu\right)$$

which implies that

$$\int (f+g) d\mu = \int f d\mu + \int g d\mu$$

■

Corollary 12.4

If $\{f_n\}$ is a sequence of non-negative measurable functions, then

$$\int \sum_{n=1}^{\infty} f_n d\mu = \sum_{n=1}^{\infty} \int f_n d\mu$$

Proof: Let $F_N = \sum_{n=1}^{N} f_n$. We see that $0 \leq F_1 \leq F_2 \leq \cdots$, so

$$\lim_{N\to\infty} \int F_N d\mu = \int \lim_{N\to\infty} F_N d\mu = \int \sum_{n=1}^{\infty} f_n d\mu$$

Then, for each N, as we have that

$$\int F_N d\mu = \sum_{n=1}^{N} \int f_n d\mu$$

by Corollary 12.3, we see that

$$\sum_{n=1}^{\infty} \int f_n d\mu = \int \sum_{n=1}^{\infty} f_n d\mu$$

■

Finally, we will state a lemma which we will prove next time.

Lemma 12.5: Fatou's Lemma

Given a sequence $\{f_n\}$ of non-negative measurable functions, we have that

$$\int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu$$

This lemma actually lets us understand our previous examples. Let $f_n = \chi_{[n,n+1]}$. We see that $\liminf_{n\to\infty} f_n = 0$, so clearly

$$\int \liminf_{n\to\infty} f_n d\mu \leq 1 = \liminf_{n\to\infty} \int f_n d\mu$$

9/25/26 Lecture 13: Dominated Convergence Theorem

First, we will give a proof of Fatou's Lemma, as promised last section.

Lemma 13.1: Fatou's Lemma

For any sequence $\{f_n\}_{n=1}^{\infty}$ of non-negative measurable functions, we have that

$$\int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu$$

Proof: Consider the sequence $g_n(x) = \inf_{j\geq n} f_j(x)$. Notice it is a sequence of non-negative functions such that $g_n \uparrow \liminf_{n\to\infty} f_n$. Now, as $g_n \leq f_j$ for all $j \geq n$, we see that

$$\int g_n d\mu \leq \inf_{j\neq n} \int f_j d\mu$$

Thus, taking limits, we see that

$$\liminf_{n\to\infty} \inf f_n d\mu \geq \lim_{n\to\infty} \int g_n d\mu = \int \liminf_{n\to\infty} f_n d\mu$$

by the MCT.

■

Note that the non-negative requirement is important: If we let $f_n = -1/n\chi_{[n,2n]}$ then $\liminf_{n\to\infty} f_n = 0$, but $\int f_n d\mu = -1$ for all n.

Now, we will prove the Dominated Convergence Theorem, which tells us that we can interchange limits when our sequence of functions is dominated by an integrable function.

Theorem 13.2: Dominated Convergence Theorem

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of measurable functions, and let $f_n(x) \to f$ for all $x \in X$. Suppose that there is some non-negative integrable function g for which $|f_n(x)| \leq g(x)$ for all x. Then,

$$\int f d\mu = \lim n \to \infty \int f_n d\mu$$

Proof: The idea is that $f_n + g \geq 0$ and $g - f_n \geq 0$ are both positive functions. By Fatou's lemma, we see that

$$\int (f+g) d\mu \leq \liminf_{n\to\infty} \int (f_n + g) d\mu = \liminf_{n\to\infty} \int f_n d\mu + \int g d\mu$$

Since g is integrable, we can subtract it to get

$$\int f d\mu \leq \liminf_{n\to\infty} \int f_n d\mu$$

We also have by Fatou's lemma that

$$\int g - f d\mu \leq \liminf_{n\to\infty} \int g - f_n d\mu = \int g + \liminf_{n\to\infty} \int -f_n d\mu$$

Subtracting $\int g$ again, we have

$$-\int f d\mu \leq \liminf_{n\to\infty} -\int f_n d\mu$$

Now, recall that $\liminf_{n\to\infty} -\int f_n d\mu = -\limsup_{n\to\infty} \int f_n d\mu$, so dividing by -1, we get

$$\int f d\mu \geq \limsup_{n\to\infty} \int f_n d\mu$$

THus, we see that

$$\int f d\mu \leq \liminf_{n\to\infty} \int f_n d\mu \leq \limsup_{n\to\infty} \int f_n d\mu \leq \int f d\mu$$

so equality holds and thus

$$\int f d\mu = \lim_{n\to\infty} \int f_n d\mu$$

■

Now, here's a corollary of the MCT and DCT.

Corollary 13.3

If f_n is a sequence of functions satisfying the hypotheses of the monotone or dominated convergence theorems, but suppose that $f_n(x) \to f(x)$ a.e.. Then, we still have that

$$\int \lim_{n\to\infty} d\mu = \int \lim_{n\to\infty} f_n d\mu$$

Proof: Let A be the set of points for which $f_n(x)$ converges to $f(x)$. Notice that for any measurable function, we must have that

$$\int g d\mu = \int g\chi_A d\mu + \int g\chi_{A^c} d\mu = \int g\chi_A d\mu$$

because the integral of a function only supported on a measure 0 set is 0.

Then, applying this fact to $f_n\chi_A$, we see that

$$\lim_{n\to\infty} \int f_n d\mu = \lim_{n\to\infty} \int f_n\chi_A d\mu = \int f\chi_A d\mu = \int f d\mu$$

■

Now, some notation that we will use in the next proposition, as well as for the rest of this class. For $A \subseteq X$, define

$$\int_A f d\mu := \int f\chi_A d\mu$$

Also, if we have a Lebesgue measurable function $f : \mathbb{R} \to \mathbb{R}$, write

$$\int f(x)dx := \int f d\lambda$$

where λ is the Lebesgue measure. Finally, we define

$$\int_a^b f d\mu = \begin{cases} \int_{[a,b]} f d\mu & a < b \\ -\int_{[b,a]} f d\mu & b < a \end{cases}$$

Proposition 13.4

The following are true about integrals.

1. If $f \geq 0$ is measurable and $\int f d\mu = 0$, the $f = 0$ almost everywhere.
2. If f is integrable and $\int_A d\mu = 0$ for all measurable A, then $f = 0$ almost everywhere.
3. If $f : \mathbb{R} \to \mathbb{R}$ is Lebesgue integrable and, for some a, $\int_a^x f(y)dy = 0$ for all x, then $f = 0$ almost everywhere.

Proof:

(1) Notice that

$$f \geq \frac{1}{n} \chi_{\{x \,:\, f(x) > 1/n\}}$$

Thus, we see that

$$\int f d\mu \geq \frac{1}{n} \mu(\{x \,:\, f(x) > 1/n\})$$

which implies that $\mu(\{x \,:\, f > 1/n\}) = 0$. Thus, as $\bigcup_{n=1}^N \{x \,:\, f(x) > 1/n\} \uparrow \{x \,:\, f(x) > 0\}$, we see that

$$\mu(x \,:\, f(x) > 0) = 0$$

Thus, as f is non-negative, this implies that f is 0 almost everywhere.

(2) Apply (1) to the sets $f\chi_{\{x \,:\, f(x) > 0\}}$ and $-f\chi_{\{x \,:\, f(x) > 0\}}$, which are both non-negative.

(3) By linearity, we see that

$$\int_b^c f(y)dy = \int_a^c f(y)dy - \int_a^b f(y)dy = 0$$

for all intervals $[b, c]$. Then, this implies that for any open set $U = \bigcup_n (a_n, b_n)$ (take intervals disjoint), we have that

$$\int_U f d\mu = \int \sum_n f(y) \chi_{(a_n, b_n)}(y) dy = \sum_n \int_{a_n}^{b_n} f(y)dy = 0$$

by the dominated convergence theorem applied to the sums (the bounding function is f).

9/30/26 Lecture 14: Riemann and Lebesgue Integrals

Proof of 13.4 cont. Recall that, if $a \in [b,c]$ we had

$$\int_b^c f(y)dy = \int_b^a f(y)dy + \int_a^c f(y)dy = 0$$

for all b, c by our assumption. If $a \notin [b,c]$, then we get a similar identity that shows $\int_b^c f(y)dy = 0$.

Now, if U is open, then we can written as $U = \bigcup_{i=1}^{\infty}(b_n, c_n)$ with the intervals taken to be disjoint, which tells us that

$$\begin{aligned}\int_U f dy &= \int \sum_n f(y)\chi_{b_n,c_n} dy \\ &= \sum_n \int_{b_n}^{c_n} f(y)dy \\ &= 0\end{aligned}$$

Then, for any measurable A, then we can write $A = (\bigcap_{n=1}^{\infty} U_n) \cup E$ where U_n is open and E is a null set. Thus, we have

$$\int_A f d\mu = \int_{\bigcap_{n=1}^{\infty} U_n} f(y)dy + \int_E f dy = 0$$

as E is a null set, and as $\bigcap_{n=1}^{\infty} U_n$ is in an open set. Thus, we see that f is zero almost everywhere by 2). ■

Proposition 14.1

For any Lebesgue integrable function f, for all $\epsilon > 0$, there exists a continuous g with a compact support so that

$$\int |f - g|\, dy < \epsilon$$

Proof: Let's start with simple functions. Consider χ_A, where A is a bounded set. By inner and outer regularity, we can find compact K and open U with $K \subseteq A \subseteq U$ so that

$$\lambda(A \setminus K) < \epsilon/2 \quad \text{and} \quad \lambda(U \setminus A) < \epsilon/2$$

Then, let

$$g(x) = \frac{d(x, U^c)}{d(x,K) + d(x, U^c)}$$

This function is continuous and is 1 on K and 0 on U^c.

Now, we see that $|\chi_A - g| \leq 1$, and is supported on $(A \setminus K) \cup (U \setminus A)$, which is a set with measure zero, so

$$\int |\chi_A - g|\, dx \leq \epsilon$$

Additionally, we see that the support of g is the closure of U, which is what we want.

Now, let's do simple functions. Let f be a simple function with compact support

$$f = \sum_{j=1}^{n} a_j \chi_{A_j}$$

with A_j bounded. We can also take $a_j \neq 0$. For each j, by the previous step, we can find g_j so that

$$\int |\chi_{A_j} - g_j| < \frac{\epsilon}{n a_j}$$

Now, consider

$$g = \sum_{j=1}^{n} a_j g_j$$

then, we see that

$$\begin{aligned}
\int \left| \sum_j a_j \chi_{A_j} - g \right| dx &\leq \sum_j \int |a_j \chi_j - a_j g_j| \, dx \\
&\leq \sum_j a_j \left(\frac{\epsilon}{n a_j} \right) dx \\
&\leq \epsilon
\end{aligned}$$

Now, let f be nonnegative with compact support. By the definition of the Lebesgue integral, we can find a simple s such that $s \leq f$ and

$$\int f \leq \int s + \epsilon/2$$

Then, applying the above step to s, we can find g so that $\int |s - g| \, dx < \epsilon/2$, which gives us that

$$\begin{aligned}
\int |f - g| \, dx &\leq \int |f - s| \, dx + \int |s - g| \, dx \\
&= \int (f - s) dx + \int |s - g| \, dx \\
&< \epsilon
\end{aligned}$$

Now, let f be nonnegative but with a support that is not necessarily bounded. Now, we know that

$$f_n = f \chi_{[-n,n]} \uparrow f$$

by the MCT, we have that $\int f_n dx \uparrow \int f$. Thus, we can pick n large enough so that

$$\int f dx - \int f_n dx < \epsilon/2$$

then we can apply the previous step to the f_n (which now is nonnegative with compact support).

Finally, consider a general Lebesgue integrable f. Then, we can apply step 4 to f^+ and f^- and use the triangle inequality on $f = f^+ - f^-$.

■

Now that we've defined a new notion of integration, let's go back and see if it coincides with the Riemann

integral. Consider a bounded function $f : [a, b] \to \mathbb{R}$. If P is a partition of $[a, b]$,

$$\{x_0, \dots, x_n\} \text{ with } a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

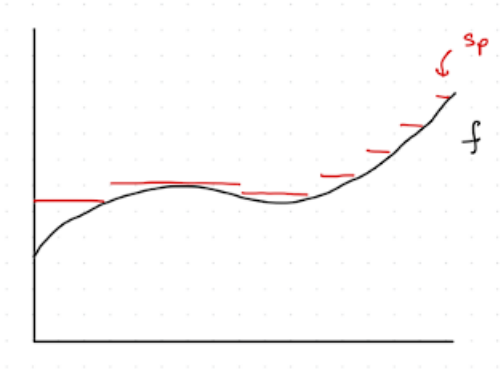
we have

$$U(P, f) = \sum_{j=1}^{n} \sup_{x \in [x_{j-1}, x_j]} f(x) \cdot \lambda[x_{j-1}, x_j]$$

$$L(P, f) = \sum_{j=1}^{n} \inf_{x \in [x_{j-1}, x_j]} f(x) \cdot \lambda[x_{j-1}, x_j]$$

When $\inf_P U(P, f) = \sup_P L(P, f)$ we say that f is Riemann integrable with the integral being either of these values. Consider the simple function

$$s_P = \sum_{i=1}^{n} (\sup_{x \in [x_{j-1}, x_j]} f(x)) \chi_{[x_{j-1}, x_j]}(x)$$



Notice that $\int s_P(x)dx = U(P, f)$ (where we are taking the Lebesgue integral).

Thus, Riemann integration is like Lebesgue integration, except for the fact that we can only use simple functions over intervals (so strictly weaker). Here are some interesting things to note that will lead us to the following theorem.

Example

Consider $\chi_{\mathbb{Q} \cap [0,1]}$. This is not Riemann integrable, but it isn't continuous anywhere (importantly, not continuous on a full measure set).

Now, consider the Thomae's function, which is 0 on the irrationals, and equal to the denominator on any rational (in lowest terms). This is Riemann integrable, and is continuous on $[0, 1] \setminus \mathbb{Q}$, a full measure set.

Theorem 14.2: Equivalence of Riemann and Lebesgue Integrals

Let $f : [a, b] \to \mathbb{R}$ be bounded.

1. If f is Riemann integrable then it is continuous Lebesgue almost everywhere and its Riemann integrable is equal to it's Lebesgue integrable.
2. If f is measurable and bounded almost everywhere, then it is Riemann integrable.

Proof: Let P be any partition and define

$$T_P = \sum_{j=1}^{n} \left(\sup_{x \in [x_{j-1}, x_j]} f(x) \right) \chi_{[x_{j-1}, x_j]}(x)$$
$$S_P = \sum_{j=1}^{n} \left(\inf_{x \in [x_{j-1}, x_j]} f(x) \right) \chi_{[x_{j-1}, x_j]}(x)$$

Let $R(f)$ denote the Riemann integral of f. By it's definition, we can find partitions Q_j and Q_j' so that

$$U(Q_j, f) \downarrow R(f) \quad \text{and} \quad L(Q', f) \uparrow R(f)$$

Now, recall that if $P \subseteq P'$, then

$$U(P', f) \leq U(P, f) \quad \text{and} \quad L(P', f) \geq L(P, f)$$

So if we define

$$P_j := \bigcup_{i \leq j} (Q_i \cup Q_i')$$

then we still have that

$$U(P_j, f) \downarrow R(f) \quad \text{and} \quad L(P_j, f) \downarrow R(f)$$

10/2/26 Lecture 15: Modes of Convergence

Proof of Theorem 14.2 (cont.) Now, recall that we have a sequence $P_1 \subseteq P_2 \subseteq \cdots$ of partitions so that T_{P_j} is decreasing and S_{P_j} is increasing. Also, we have that $T_{P_j} \geq f \geq S_{P_j}$, and by the monotone convergence theorem, there must be some T and S so that $T_{P_j} \to T$ and $S_{P_j} \to S$. Also, recall these functions have the property that

$$\int T_{P_j} dx = U(P_j, f) \quad \text{and} \quad \int S_{P_j} dx = L(P_j, f)$$

Now, $\left|T_{P_j}\right| \leq \sup_{x\in[0,1]} |f(x)| < \infty$ (f bounded) and the same inequality holds for $\left|S_{P_j}\right|$. Because of this, we can apply the DCT to the following:

$$\begin{aligned}\int T - S dx &= \lim_{j\to\infty} \int T_j - S_j dx \\ &= \lim_{j\to\infty} U(P_j, f) - L(P_j, f) \\ &= 0\end{aligned}$$

as f is Riemann integrable. Then, because $\int (T-S)dx = 0$, and because $T - S \geq 0$, we see that $T = S$ almost everywhere. Thus, we must have that $T = f = S$ almost everywhere.

Now, we want to show that f is continuous a.e., so what we want to show that if $T(x) = S(x)$ and $x \notin P_j$ for any j, then f is continuous at x. Let $\epsilon > 0$. Then, as T and S agree at x, find a j so that

$$T_{P_j}(x) - S_{P_j}(x) < \epsilon$$

and choose δ so that $x \in (x-\delta, x+\delta) \subseteq [x_{j-1}, x_j]$. On this interval, note that T_{P_j} and S_{P_j} are constant. Thus, $T_{P_j}(y) - S_{P_j}(y) < \epsilon$ for all $y \in (x-\delta, x+\delta)$. Notice this implies that $T_{P_k}(y) - S_{P_k}(y) < \epsilon$ for all $k \geq j$ on $(x-\delta, x+\delta)$, which implies that f is continuous at x (bound the variance of f by the constant values of T_{P_j} and S_{P_j}).

Finally, we see that

$$\int f = \lim_{j\to\infty} \int T_{P_j} = \lim_{j\to\infty} U(P_j, f) = R(f)$$

so the Lebesgue and Riemann integrals of f agree, which was to be shown.

(2) Let $P_1 \subseteq P_2 \subseteq \cdots$ be some increasing sequence of partitions with $\text{mesh}(P_j) \to 0$. For instance, take

$$P_j = \{a + n(b-a)2^{-j} \,:\, n = 0, \ldots, 2^j\}$$

Suppose that x is a point of continuity of f that is not in any of the P_j. For each j, pick $[y_j, z_j]$ (with endpoints in P_j) containing x. Now $y_j \to x$ and $z_j \to x$. Then, we see that

$$\sup_{y\in[y_j,z_j]} f(y) \to f(x)$$

because f is continuous as x. Thus, we see that $T_{P_j}(x) \to f(x)$ and by the same argument, $S_{P_j}(x) \to f(x)$ for almost every $x \in [a,b]$. Then,

$$U(P_j, f) = \int T_{P_j} dx \to \int f dx$$

and

$$L(P_j, f) = \int S_{P_j} dx \to \int f dx$$

Thus, we see that f is Riemann integrable.

Modes of Convergence

Let $(X, \mathcal{A}, \mu)$ be a measure space.

Definition 15.1: Pointwise Almost Everywhere Convergence

Given $f_n : X \to \mathbb{R}$ measurable, we say that $f_n \to f$ *pointwise almost everywhere* if

$$\mu\{x : f_n(x) \not\to f(x)\} = 0$$

We usually denote this $f_n \to f$ a.e.

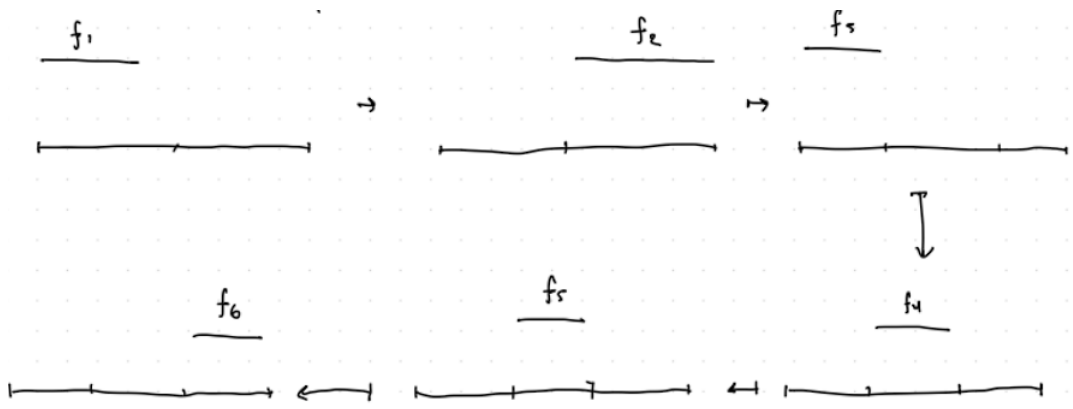
Definition 15.2: Convergence in Measure

Given $f_n : X \to \mathbb{R}$ measureable, we say that $f_n \to f$ *in measure* if, for any $\epsilon > 0$,

$$\mu\{x : |f_n(x) - f(x)| > \epsilon\} \to 0$$

as $n \to \infty$.

It turns out convergence in measure does not imply convergence pointwise almost everywhere. Consider the following function:



Note that if we go far enough, $\mu(|f_n| > 0)$ is $1/n$, so f_n converges in measure to 0. However, notice that $f_n(x) = 1$ infinitely many times, so f_n never converges pointwise to 0.

It also turns out that, when $\mu(X) = \infty$, convergence pointwise does not imply convergence in measure. Consider the functions

$$f_n = \chi_{[n,n+1]}$$

We see that $f_n \to 0$ almost everywhere (pointwise), but $\mu(|f_n|(x) > 0) = 1$. Convergence pointwise seems stronger, but we have problems when mass escapes to infinity. Thus, if we assume this isn't possible, it turns out we can have one implication.

Proposition 15.3

Let $f_n : X \to \mathbb{R}$ be measurable.

1. If μ is a finite measure, then $f_n \to f$ almost everywhere implies that $f_n \to f$ in measure.
2. If μ is any measure, and if $f_n \to f$ in measure, then there is a subsequence $f_{n_j} \to f$ almost everywhere.

Proof **(1)** Let

$$A_n = \{x : |f_n(x) - f(x)| > \epsilon\}$$

Then, as $f_n(x) \to f(x)$ almost everywhere, this implies that $\chi_{A_n}(x) \to 0$ almost everywhere. Then, we see that

$$\lim_{n\to\infty} \mu(A_n) = \int \lim_{n\to\infty} \chi_{A_n}(x)dx = \int 0 = 0$$

where the dominated convergence theorem was needed for the above equality, using the fact that $|\chi_{A_n}| \leq 1$ and 1 is integrable when $\mu(X) < \infty$.

10/5/26 Lecture 16: L^p Convergence and Product Measures

Proof of (2): Let $n_1 = 1$. Now, pick $n_j > n_{j-1}$ so that

$$\mu(A_j) < 2^{-j}$$

where $A_j = \{x : \left|f_{n_j}(x) - f(x)\right| > 1/j\}$. Let

$$A = \bigcap_{k=1}^{\infty} \bigcup_{j \geq k} A_j$$

so we see

$$\begin{aligned}\mu(A) &= \lim_{k\to\infty} \mu\left(\bigcup_{j\geq k} A_j\right) \\ &= \lim_{k\to\infty} \sum_{j=k}^{\infty} 2^{-j} = 0\end{aligned}$$

We claim that if $x \notin A$, then $f_{n_j}(x) \to f(x)$. Note that points in A are points that are in infinitely many A_j. Thus, for sufficiently large j, we see that $\left|f_{n_j}(x) - f(x)\right| < \frac{1}{j}$, so $f_{n_j}(x) \to f(x)$. Thus, f_{n_j} converges almost everywhere to f.

■

Now, we give one more definition of convergence.

Definition 16.1: Convergesnce in L^p

Let $1 \leq p < \infty$. A sequence of measurable functions f_n *converges to* f *in* L^p if

$$\int |f_n - f|^p \, d\mu \to 0$$

as $n \to \infty$.

Why didn't we just choose $p = 0$? Here is why:

Example

Let $f_n = n^{1/2}\chi_{[0,1/n]}$. We see that

$$\int |f_n - 0| \, dx = \frac{1}{n^{1/2}} \to 0$$

so $f_n \to 0$ in L^1. However, notice that

$$\int |f_n - 0|^2 \, dx = \int n\chi^2_{[0,1/n]} dx = \int 1 dx = 1$$

Heuristically, in L^1 area under the curve is width times height, but in L^2 area is width times height squared.

Lemma 16.2: Chebyshev's Inequality

For any integrable f and some $1 \leq p < \infty$, then for any $\epsilon > 0$,

$$\mu(\{x : |f(x)| \geq a\}) \leq \frac{\int |f|^p \, d\mu}{a^p}$$

Proof: Notice that

$$a^p \chi_{\{x : |f(x)| \geq a\}} \leq |f(x)|^p$$

on A. Integrating, we see that

$$a^p \mu(\{x : |f(x)| \geq a\}) \leq \int_A |f|^p \leq \int |f|^p$$

which gives us our desired inequality.

■

Proposition 16.3

Convergence in L^p implies convergence in measure, which implies a subsequence converges almost everywhere.

Proof: Fix $\epsilon > 0$. By Chebyshev's inequality, we have that

$$\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) \leq \epsilon^p \int |f_n(x) - f(x)|^p \, d\mu$$

which goes to zero as $n \to \infty$.

■

Now, recall that $f_n(x) \to f$ uniformly on A if

$$\lim_{n \to \infty} \sup_{x \in A} |f_n(x) - f(x)| = 0$$

This brings us to a very useful theorem.

Theorem 16.4: Egorov's Theorem

Let μ be a finite measure. Then, if $f_n \to f$ almost everywhere, for any $\epsilon > 0$, there exists a set A with $\mu(A) < \epsilon$ such that $f_n \to f$ uniformly on A^c.

Proof: Define the sets

$$A_{n,k} = \bigcup_{m \geq n} \{x : |f_m(x) - f(x)| > 1/k\}$$

Fixing k, we see that $A_{n,k}$ is decreasing in n. Now, notice that $\bigcap_n A_{n,k} \subseteq \{x : f_m(x) \not\to f(x)\}$, which was measure zero. We also see that

$$\lim_{n \to \infty} \mu(A_{n,k}) = \mu\left(\bigcap_n A_{n,k}\right)$$

which goes to zero, so choose n_k so that $\mu(A_{n_k,k}) < \epsilon 2^{-k}$. Now, let

$$A = \bigcup_{k=1}^{\infty} A_{n_k,k}$$

by subadditivity, $\mu(A) < \epsilon$. Now, if $x \notin A$, then for all k,

$$|f_n(x) - f(x)| < 1/k$$

for $n \geq n_k$. As this holds for all $x \in A^c$, we see that

$$\sup_{x \in A} |f_n(x) - f(x)| \leq 1/k$$

for all $n \geq n_k$, so uniform convergence follows taking $k \to 0$.

■

Remark: This can be used to give a pretty quick proof of Lusin's theorem.

Product Measures

Recall the Lebesgue measure on $\mathbb{R}$. The way we constructed it was starting with a premeasure on algebra of intervals, moving this to the Lebesgue outer measure, and then going to the measure. Now what we want is the Lebesgue measure on $\mathbb{R}^n$.

Lets first consider $\mathbb{R}^2$. Define

$$\lambda^2([a,b] \times [c,d]) = (b-a) \cdot (d-c) = \lambda([a,b]) \cdot \lambda([c,d])$$

Thus, we are thinking of $\mathbb{R}^2$ as $\mathbb{R} \times \mathbb{R}$, and saying that the measures of rectangles is length times height!

Fix two measure spaces $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$.

Definition 16.5: Measurable Rectangles and Product σ-Algebras

A *measurable rectangle* in $X \times Y$ is a set of the form $A \times B$ where $A \in \mathcal{A}$ and $B \in \mathcal{B}$. The *product σ-algebra* is the σ-algebra generated by measurable rectangles.

Now, we want to define a premeasure on an algebra that we can then extend. Let $\mathcal{C}_0$ be the collection of finite unions of measurable rectangles. We claim that this is an algebra. It is clear this is closed under finite unions, so we need to check complements. We see that

$$(A \times B)^c = (X \times B^c) \cup (A^c \times B)$$

Now, for $A \times B$, let $\ell(A \times B) = \mu(A) \cdot \nu(B)$.

Proposition 16.6

Let $A \times B = \bigcup_{n=1}^{\infty} A_n \times B_n$ where $A_n \in \mathcal{A}$, $B_n \in \mathcal{B}$, and they are all disjoint. Then,

$$\ell(A \times B) = \sum_n \ell(A_n \times B_n)$$

which is to show that

$$\mu(A)\nu(B) = \sum_n \mu(A_n)\nu(B_n)$$

Proof: First, write

$$\chi_A(x)\chi_B(y) = \chi_{A\times B}(x,y) = \sum_n \chi_{A_n \times B_n}(x,y) = \sum_n \chi_{A_n}(x)\chi_{B_n}(y)$$

fixing y, we get an equality of functions of x, so integrating

$$\begin{aligned}\mu(A)\chi_B(y) &= \sum_n \int \chi_{A_n}(x)\chi_{B_n}(y)d\mu(x) \\ &= \sum_n \mu(A_n)\chi_{B_n}(x)\end{aligned}$$

Then, as this gives us a function of y, we see that integrating this again,

$$\mu(A)\mu(B) = \sum_n \mu(A_n) \int \chi_{B_n}(y)d\nu = \sum_n \mu(A_n)\nu(B_n)$$

■

By the Caratheodory extension theorem, we get a measure on $(X \times Y, \mathcal{A} \times \mathcal{B})$ with the property that

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B), \quad A \in \mathcal{A}, B \in \mathcal{B}$$

10/7/26 Lecture 17: Product Measures (cont.)

Recall from last time we defined a measure on $(X\times, Y, \mathcal{A}\times\mathcal{B})$ with the property that

$$(\mu\times\nu)(A\times B)=\mu(A)\times\nu(B)$$

and by Caratheodory, this is unique. Now, we need to define how to integrate with respect to this measure.

Let $E\subseteq X\times Y$, and, for fixed x, y, define

$$E_x=\{y\in Y\ :\ (x,y)\in E\}$$

and

$$E^y=\{x\in X\ :\ (x,y)\in E\}$$

Similarly, given a function $f: X\times Y\to\mathbb{R}$, for fixed x, y, define

$$f_x: Y\to\mathbb{R} \text{ by } f_x(y)=f(x,y)$$

and

$$f^y: X\to\mathbb{R} \text{ by } f^y(x)=f(x,y)$$

Proposition 17.1

1. If $E\in A\times B$, then $E_x\in\mathcal{B}$ for any $x\in X$ and $E^y\in\mathcal{A}$ for any $y\in Y$.
2. If f is $\mu\times\nu$ measurable, then f_x and f^y are measurable with respect to μ and ν.

Proof:
(1) Let $\mathcal{C}$ be the collection of subsets $E\subseteq X\times Y$ such that $E_x\in\mathcal{B}$ for all x and $E^y\in\mathcal{A}$ for all y. We want to show that $\mathcal{C}$ contains measurable rectangles and is a σ-algebra.

First, we see that if $E=A\times B$, then

$$E_x=\begin{cases}B & x\in A\\ \emptyset & x\notin A\end{cases}$$

and

$$E^y=\begin{cases}A & y\in B\\ \emptyset & y\notin B\end{cases}$$

so all rectangles are in $\mathcal{C}$. Clearly, $\mathcal{C}$ is a σ-algebra as it is clear that it is closed under unions and complements (pass conditions to the restrictions). Thus, we see that $\mathcal{A}\times\mathcal{B}\in\mathcal{C}$, which was to be shown.

(2) We see that for $U\subseteq\mathbb{R}$, so as

$$f_x^{-1}(U)=(f^{-1}(U))_x$$

so if $f^{-1}(U)\in\mathcal{A}\times\mathcal{B}$, we see that $f_x^{-1}(U)\in\mathcal{B}$ by (1).

■

Definition 17.2: Monotone Classes

We say a set $\mathcal{M}\subseteq\mathcal{P}(X)$ is a *monotone class* if

1. $A_1\subseteq A_2\subseteq\cdots,\quad A_n\in\mathcal{M}\implies\bigcup_n A_n\in\mathcal{M}$
2. $A_1\supseteq A_2\supseteq\cdots,\quad A_n\in\mathcal{M}\implies\bigcap_n A_n\in\mathcal{M}$

Given a collection $\mathcal{C} \in \mathcal{P}(X)$, the monotone class generated by $\mathcal{C}$ is the smallest monotone class containing $\mathcal{C}$; that is, it is the intersection of all monotone classes containing $\mathcal{C}$.

Theorem 17.3: Monotone Class Theorem

Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra. Let $\mathcal{M}$ be the monotone class generated by $\mathcal{A}$. Then, $M = \sigma(\mathcal{A})$.

Proof: Firstly, it is clear that $\mathcal{M} \subseteq \sigma(\mathcal{A})$ as σ-algebras are monotone classes. Now, we want to show that $\sigma(\mathcal{A}) \subseteq \mathcal{M}$, which is to show that $\mathcal{M}$ is a σ-algebra.

Firstly, we need to show that if $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$. Let $\mathcal{N}_1 = \{A \in \mathcal{M} : A^c \in \mathcal{M}\}$. We see that $\mathcal{N}_1$ contains $\mathcal{A}$. Now, if $A_1 \subseteq A_2 \subseteq \cdots$ with $A_n \in \mathcal{N}_1$, since $\bigcup_n A_n \in \mathcal{M}$, and

$$\left(\bigcup_n A_n\right)^c = \left(\bigcap_n A_n^c\right)$$

then as $A_1^c \supseteq A_2^c \supseteq \cdots$, we see that $\bigcap_n A_n^c = (\bigcup_n A_n)^c \in \mathcal{M}$. Thus, $\bigcup_n A_n \in \mathcal{N}_1$. By the same argument, we see that if $A_1 \supseteq A_2 \supseteq \cdots$, then $\bigcap_n A_n \in \mathcal{N}_1$, so $\mathcal{N}_1$ is a monotone class. Thus, $\mathcal{M} \subseteq \mathcal{N}_1$, so $\mathcal{M}$ is closed under complements.

Now, we want to show that it is closed under intersection. Start by letting $\mathcal{N}_2 = \{B \in \mathcal{M} : A \cap B \in \mathcal{M} \text{ for all } A \in \mathcal{A}\}$. We see that $\mathcal{N}_2$ contains $\mathcal{A}$. Now, suppose that $B_1 \subseteq B_2 \subseteq \cdots$ with $B_n \in \mathcal{N}_2$. Then, for any $A \in \mathcal{A}$, we see that as $\bigcup_n B_n \in \mathcal{M}$,

$$\bigcup_n B_n \cap A = \bigcup_n (B_n \cap A)$$

and as $B_1 \cap A \subseteq B_2 \cap A \cdots$ and $B_n \cap A \in \mathcal{M}$, we see that $\bigcup_n B_n \cap A \in \mathcal{M}$, so $\bigcup_n B_n \in \mathcal{N}_2$. Since a similar proof for intersections follows, we see that $\mathcal{M} \subseteq \mathcal{N}_2$.

For the next step, let $\mathcal{N}_3 = \{B \in \mathcal{M} : A \cap B \in \mathcal{M} \text{ for all } A \in \mathcal{M}\}$. Now, $\mathcal{N}_3$ contains $\mathcal{A}$ because of the last step. By the same argument as above, we see that $\mathcal{N}_3$ is a monotone class and so $\mathcal{M} \subseteq \mathcal{N}_3$.

The last thing to check is countable intersections. Let $\{A_n\} \subseteq \mathcal{M}$. Now, since we know that $\bigcap_{i=1}^n A_i \in \mathcal{M}$ for each n, and as

$$\bigcap_{i=1}^1 A_i \supseteq \bigcap_{i=1}^2 A_i \supseteq \cdots$$

we see that

$$\bigcap_{n=1}^\infty \bigcap_{i=1}^n A_i = \bigcap_{i=1}^\infty A_n \in \mathcal{M}$$

Thus, $\mathcal{M}$ is a σ-algebra, which was to be proved.

■

Proposition 17.4

Suppose that μ and ν are σ-finite. If $E \in \mathcal{A} \times \mathcal{B}$, then $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y respectviely, and

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu = \int_Y \mu(E^y) dy$$

Proof: First, assume that μ and ν are finite. Then, let $\mathcal{M}$ be the collection of the sets so that the conclusions of the theorem holds. We want to show that $\mathcal{M}$ contains finite unions of measurable rectangles and is a monotone class.

First, let $E = A \times B$. Then, as

$$E_x = \begin{cases} \emptyset & x \notin A \\ B & x \in A \end{cases}$$

so $\nu(E_x) = \chi_A(x) \cdot \nu(B)$, which is measurable, and similarly for E^y. Similarly, we also see that

$$\int_X \nu(E_x)dx = \mu(A)\nu(B) = (\mu \times \nu)(E)$$

and similarly for the other integral. Thus, measurable rectangles are in $\mathcal{M}$.

Now, let $E_1 \subseteq E_2 \subseteq \cdots$ are in $\mathcal{M}$ and let $E = \bigcup_n E_n$. Let $f_n(y) = \mu(E_n^y)$. Each of these are measurable, and they increase pointwise to $f(y) = \mu(E^y)$. Thus, f is measurable as it is the limit of measurable functions. Then, by the monotone convergence theorem

$$\begin{aligned} \int_Y f(y)d\nu &= \lim_{n\to\infty} \int f_n(y)d\nu \\ &= \lim_{n\to\infty} (\mu \times \nu)(E_n) \\ &= (\mu \times \nu)(E) \end{aligned}$$

To see that $E_1 \supseteq E_2 \supseteq \cdots$ implies $E = \bigcap_n E_n$, it is the same proof but we use DCT using the fact ath $\mu \times \nu$ is σ-finite.

■

10/9/26 Lecture 18: Fubini-Tonelli

Consider $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$, and consider $\mathcal{C}_0$, the algebra of rectangles $\bigcup_{i=1}^N A_i \times B_i$. Recall that for $E \subseteq X \times Y$,

$$E_x = \{y : (x, y) \in E\} \quad \text{and} \quad E^y = \{x : (x, y) \in E\}$$

and for $f : X \times Y \to \mathbb{R}$,

$$f_x(y) = f(x, y) \quad \text{and} \quad f^y(x) = f(x, y)$$

Theorem 18.1: Fubini-Tonelli

Suppose μ and ν are σ-finite and f is $\mu \times \nu$ measurable. Then, if f is nonnegative or integrable, then

$$g(x) := \int f(x, y) d\nu(y) \quad \text{and} \quad h(y) := \int f(x, y) d\mu(x)$$

are measurable and

$$\begin{aligned} \int f(x, y) d(\mu \times \nu)(x, y) &= \iint f(x, y) d\nu(y) d\mu(x) \\ &= \iint f(x, y) d\mu(x) d\nu(y) \end{aligned}$$

Proof: By Proposition 17.4 from last time, the theorem holds for simple functions. Now, if f is non-negative, let $0 \leq f_n \uparrow f$ be a sequence of simple functions increasing to f. Let

$$g_n(x) := \int f_n(x, y) d\nu(y) \quad \text{and} \quad h_n(y) := \int f_n(x, y) d\mu(x)$$

We see that as $f_n(x, y) \uparrow f(x, y)$, $g_n(x) \uparrow g(x)$ and $h_n(y) \uparrow h(y)$ by the MCT. Thus, g and h are measurable.

Now,

$$\int g(x) d\mu(x) = \lim_{n \to \infty} \int g_n(x) d\mu(x) = \lim_{n \to \infty} \int f_n(x, y) d(\mu \times \nu)(x, y) = \int f(x, y) d(\mu \times \nu)(x, y)$$

by the MCT. The analogous result for h holds.

Now, if f is integrable (this is so that $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$), write $f = f^+ - f^-$ and apply the nonnegative result.

■

Remark

The utility of the theorem is that we can switch the order of integration:

$$\int g(x) d\mu(x) = \int h(y) d\nu(y)$$

Example

Let $A \subseteq \mathbb{R}$ have finite Lebesgue measure and let $f(t) = \lambda(A \cap (A+t))$. We want to show that

$$\int f(t)dt = (\lambda(A))^2$$

Proof: We can write

$$\int f(t)dt = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_A(x)\chi_{A+t}(x)dx \right) dt$$

Since $\chi_A(x)\chi_{A+t}(x) \geq 0$, we can switch the order of integration to get

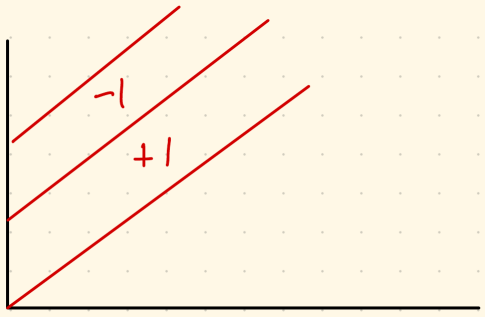
$$\begin{aligned}
\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_A(x)\chi_{A+t}(x)dx \right) dt &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_A(x)\chi_{A+t}(x)dt \right) dx \\
&= \int_{\mathbb{R}} \chi_A(x) \left(\int_{\mathbb{R}} \chi_{x-A}(t)dt \right) dx \\
&= \int_{\mathbb{R}} \chi_A(x)\lambda(A)dx \\
&= \lambda(A)^2
\end{aligned}$$

where we used that $\chi_{A+t}(x) = \chi_{x-A}(t)$.

Example: Necessity of Integrability

Let $f : [0,\infty)^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} 1 & x \leq y < x+1 \\ -1 & x+1 \leq y \leq x+2 \\ 0 & \text{else} \end{cases}$$



Then,

$$\int_0^\infty f(x,y)dy = 0 \quad \text{and} \quad \int_0^\infty f(x,y)dx = \begin{cases} 0 & y \geq 2 \\ 2-y & 1 \leq y \leq 2 \\ y & 0 \leq y \leq 1 \end{cases}$$

so

$$\int_0^\infty \int_0^\infty f(x,y)dydx = 0 \quad \text{and} \quad \int_0^\infty \int_0^\infty f(x,y)dxdy = 1$$

Luckily, f is not integrable, as $\int f^+ = \infty$ and $\int f^- = \infty$.

10/12/26 Lecture 19: Completions and Signed Measures

Recall the product measure from last time. It is quite clear how to extend it to finitely many measure spaces by

$$(\mu_1 \times \cdots \times \mu_n)(A_1 \times \cdots \times A_n) = \prod_{i=1}^{n} \mu_i(A_i)$$

Unfortunately, product measures are not complete. Let $A \in \mathcal{A}_1$ with $A \neq \emptyset$ and $\mu_1(A) = 0$. Take a non-measurable $E \in \mathcal{P}(A_2)$. Now, $A \times E$ is non-measurable (otherwise, E would be measurable by taking restrictions), but $A \times E \subseteq A \times A_2$, and $A \times A_2$ has measure 0. Thus, $\mathcal{A}_1 \times \mathcal{A}_2$ is not complete.

Recall that, given a measure space $(X, \mathcal{A}, \mu)$ that is not complete, we can consider its completion $(X, \overline{\mathcal{A}}, \overline{\mu})$.

Definition 19.1: d-Dimensional Lebesgue Measure

The *d-dimensional Lebesgue measure* is the completion of $(\mathbb{R}^d, \mathcal{A}^d, \lambda^d)$.

Note: We can also define the Lebesgue measure on $\mathbb{R}^d$ using an outer measure with rectangles, and it agrees!

Signed Measures

Motivation: If f is smooth and compactly supported, then

$$f(b) - f(a) = \int_a^b f'(x)dx$$

Now, as seen in the homework, $\nu_{f'}(E) = \int_E f'(x)dx$ is a measure when $f' \geq 0$, so $f(b) - f(a) = \nu_{f'}([a, b])$. This is the Lebesgue-Stieltjes measure for f! Thus, the measure acts like f' in the sense that if we integrate, we get f (even if f' doesn't exist!)

Issue: This only works for increasing, right-continuous functions, so if we want to be more general, we want to allow signed measures.

Definition 19.2: Signed Measures

Given a σ-algebra $\mathcal{A}$, a *signed measure* $\mu : \mathcal{A} \to (-\infty, \infty]$ is a function such that

1. $\mu(\emptyset) = 0$
2. If $\{A_i\}_{i=1}^{\infty}$ are disjoint, then
$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$
and the sum converges absolutely when the LHS is finite.

Proposition 19.3

Let μ be a signed measure on $(X, \mathcal{A})$. Then,

1. If $A_1 \subseteq A_2 \subseteq \cdots$ with $A_n \in \mathcal{A}$, then $\mu\left(\bigcup_n A_n\right) = \lim_n \mu(A_n)$.
2. If $A_1 \supseteq A_2 \supseteq \cdots$ with $A_n \in \mathcal{A}$ and $\mu(A_1) < \infty$, then $\mu\left(\bigcap_n A_n\right) = \lim_n \mu(A_n)$.

Examples: Signed Measures

- If μ_1 and μ_2 are measures and μ_2 is finite, then $\nu = \mu_1 - \mu_2$ is a signed measure.
- If f is integrable, then $\nu(E) = \int_E f dx$ is a signed measure. Note that this is the same as above if we let $\mu_1(E) = \int_E f^+ dx$ and $\mu_2(E) = \int_E f^- dx$.

Definition 19.4: Positive, Negative, and Null Sets

Let μ be a signed measure. A set $A \in \mathcal{A}$ is *positive* if $\mu(B) \geq 0$ for any measurable $B \subseteq A$. Similarly, A is *negative* if $\mu(B) \leq 0$ for any measurable $B \subseteq A$. Finally, A is *null* if $\mu(B) = 0$ for any measurable $B \subseteq A$.

Theorem 19.5: Hahn Decomposition Theorem

Let μ be a signed measure on $(X, \mathcal{A})$.

1. There exist disjoint sets $A, B \in \mathcal{A}$ such that A is positive, B is negative, and $A \cup B = X$.
2. If A' and B' are another pair of sets satisfying 1, then $A \triangle A'$ and $B \triangle B'$ are null sets.
3. If μ is not a positive measure, then $\mu(B) < 0$, and if $-\mu$ is not positive, then $\mu(A) > 0$.

Proof: We need a lemma first.

Lemma 19.6

Let μ be a signed measure. If $\mu(B) < 0$, then we can find $B' \subseteq B$ that is negative such that $\mu(B') < 0$.

Proof: Suppose B is not negative (the result is obvious if it is). The idea is that we pull positive pieces off of B.

As B is not negative, find the smallest integer n_1 such that there exists $A_1 \subseteq B$ with $\mu(A_1) \geq 1/n_1$. Now, let

$$B_1 = B \setminus A_1$$

If B_1 is negative, we are done, as $\mu(B_1) = \mu(B) - \mu(A_1) < 0$. If not, iterate. Given B_j not negative, find $A_{j+1} \subseteq B_j$ such that $\mu(A_{j+1}) \geq 1/n_{j+1}$, where n_{j+1} is the smallest such integer. Then, define

$$B_{j+1} = B_j \setminus A_{j+1}$$

Now, let $B' = \bigcap_j B_j$. Then, $B = B' \cup \left(\bigcup_{j=1}^{\infty} A_j\right)$. Since this is a disjoint union,

$$\mu(B) = \mu(B') + \sum_{i=1}^{\infty} \mu(A_i)$$

Since $\mu(B) < 0$, we must have $\mu(B') < 0$. Also, as $\mu(B') < 0$, $\mu(A_j) \to 0$, which implies $n_j \to \infty$.

Now, we need to check that B' is a negative set.

10/14/26 Lecture 20: The Hahn and Jordan Decompositions

Proof of Lemma 19.6 (cont.): Now, we need to check that B' is a negative set. Suppose not. Then, there exists $P \subseteq B'$ such that $\mu(P) > 0$. Then, there exists N such that $0 \leq 1/N < \mu(P)$. Now, since $n_j \to \infty$, there exists J such that $1/n_J < 1/N$. However, this is a contradiction, as n_J was chosen to be the smallest such integer for a set in B_{J-1}, but there exists a set in B_{J-1} with measure greater than $1/N$. ■

Now, we can prove the Hahn Decomposition theorem.

Theorem 20.1: Hahn Decomposition Theorem

Let μ be a signed measure on $(X, \mathcal{A})$.

1. There exist disjoint sets $A, B \in \mathcal{A}$ such that A is positive, B is negative, and $A \cup B = X$.
2. If A' and B' are another pair of sets satisfying 1, then $A \triangle A'$ and $B \triangle B'$ are null sets.
3. If μ is not a positive measure, then $\mu(B) < 0$, and if $-\mu$ is not positive, then $\mu(A) > 0$.

Proof: **(1)** Let

$$L = \inf\{\mu(B) : B \text{ is a negative set}\}$$

Choose B_n so that $\mu(B_n) \downarrow L$. Now, let $B_1' = B_1$ and $B_n' = B_n \setminus \bigcup_{j=1}^{n-1} B_j'$. Now, $B = \bigcup_n B_n = \bigsqcup_n B_n'$. Then, for $E \subseteq B$, .

$$\mu(E) = \lim_{n\to\infty} \mu\left(E \cap \bigcup_{j=1}^{n} B_j'\right) = \lim_{n\to\infty} \sum_{j=1}^{n} \mu(E \cap B_j') \leq 0$$

so B is negative. Now, because B is negative,

$$\mu(B) = \mu(B_n) + \mu(B \setminus B_n) \leq \mu(B_n)$$

so taking the limit, we see that $\mu(B) \leq L$, so as $\mu(B) \geq L$, we have equality. Note that this also shows that this shows $L > -\infty$ as $\mu(B) > -\infty$.

Let $A = B^c$. We want to show that A is positive. If not, there exists $E \subseteq A$ such that $\mu(E) < 0$, so we can also find a negative $E' \subseteq E$ such that $\mu(E') < 0$. But then

$$\mu(B \cup E') = \mu(B) + \mu(E') < L$$

which is a contradiction, as $B \cup E'$ is negative.

(2) Let $E \subseteq A \setminus A' = B' \setminus B$. Then, as A is positive and B' is negative, $\mu(E) \geq 0$ and $\mu(E) \leq 0$. A similar fact holds for $E \subseteq A' \setminus A = B \setminus B'$. Thus, $A \triangle A'$ and $B \triangle B'$ are null.
(3) Suppose μ is not positive. Then, there exists E such that $\mu(E) < 0$. Then,

$$\begin{aligned} \mu(E) &= \mu(E \cap A) + \mu(E \cap B) \\ &\geq \mu(E \cap B) \\ &\geq \mu(B) && (B \text{ is negative}) \end{aligned}$$

Thus, $0 > \mu(E) \geq \mu(B)$, so $\mu(B) < 0$. Similarly, if μ is not negative, there exists E such that $\mu(E) > 0$, so

$$\begin{aligned}\mu(E) &= \mu(E \cap A) + \mu(E \cap B) \\ &\leq \mu(E \cap A) \leq \mu(A)\end{aligned}$$

so $0 < \mu(E) \leq \mu(A)$.

■

Definition 20.2: Mutually Singular Measures

Two measures μ and ν, both on $(X, \mathcal{A})$, are *mutually singular* if there exist disjoint A, B such that $A \cup B = X$ and $\mu(B) = \nu(A) = 0$. We denote this by $\mu \perp \nu$.

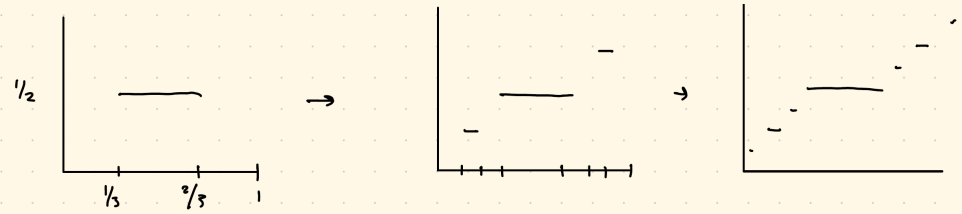
Example 1

On $\mathbb{R}$, consider the Lebesgue measure and the point mass at 0. Then, $\mathbb{R} = (\mathbb{R} \setminus \{0\}) \sqcup \{0\}$ and

$$\delta_0(\mathbb{R} \setminus \{0\}) = \lambda(\{0\}) = 0$$

Example 2

Recall the Cantor function f:



Let m be the Lebesgue-Stieltjes measure of f. We claim that λ and m are mutually singular. The idea is that m only assigns measure to the Cantor set C. Then,

$$m(C^c) = \lambda(C) = 0$$

Theorem 20.3: Jordan Decomposition Theorem

If μ is a signed measure, then there are positive measures μ^+ and μ^-, with μ^- finite, so that $\mu = \mu^+ - \mu^-$ and $\mu^+ \perp \mu^-$, and these measures are unique.

Proof: Apply the Hahn decomposition to get a positive set A and a negative set B. Then, let

$$\mu^+(E) = \mu(A \cap E) \quad \text{and} \quad \mu^-(E) = -\mu(E \cap B)$$

These are positive measures, and

$$\mu(E) = \mu(A \cap E) + \mu(E \cap B) = \mu^+(E) - \mu^-(E)$$

so $\mu = \mu^+ - \mu^-$. It is clear that $\mu^+ \perp \mu^-$ as $\mu^-(A) = \mu^+(B) = 0$ and $A \cup B = X$. We also see that μ^- is finite as $\mu^-(X) = -\mu(B) < \infty$.

Now, we check uniqueness. Suppose $\mu = \nu^+ - \nu^-$. We want to show that $\nu^+ = \mu^+$ and $\nu^- = \mu^-$. We know $\nu^+ \perp \nu^-$, so pick A' and B' so that $\nu^+(B') = \nu^-(A') = 0$. Then, for $E \subseteq A'$,

$$\mu(E) = \nu^+(E) \geq 0$$

so A' is positive for μ, and similarly B' is negative for μ. Thus, A' and B' form a Hahn decomposition for μ, so $A \triangle A'$ is null. Now,

$$\begin{aligned} \nu^+(E) = \nu^+(E \cap A') &= \nu^+(E \cap A') - \nu^-(E \cap A') \\ &= \mu(E \cap A') \\ &= \mu(E \cap A) \\ &= \mu^*(E) \end{aligned}$$

so $\mu^+ = \nu^+$. Then, as $\nu^- = \nu^+ - \mu = \mu^+ - \mu = \mu^-$, we are done.

■

10/16/26 Lecture 21: The Radon-Nikodym Theorem

If f is integrable with respect to μ, we can define

$$\nu(A) = \int_A f d\mu$$

In a sense, "f is the derivative of ν with respect to μ".

Question: When does ν have a derivative with respect to μ?

Answer: Note that $\mu(A) = 0 \implies \nu(A) = 0$. This is necessary, but it turns out it is sufficient as well.

Definition 21.1: Absolute Continuity

We say that ν is *absolutely continuous with respect to* μ if $\nu(A) = 0$ whenever $\mu(A) = 0$, denoted $\nu \ll \mu$.

Proposition 21.2

If $\nu \ll \mu$ and $\nu \perp \mu$, then $\nu = 0$.

Proof: Find A, B such that $\nu(A) = \mu(B) = 0$. Then, $\nu(B) = 0$. Thus, as $\nu(A) + \nu(B) = \nu(X)$, we see that $\nu(X) = 0$.

■

Proposition 21.3

Let ν be a finite measure. Then, $\nu \ll \mu$ if, and only if, for any $\epsilon > 0$, there exists $\delta > 0$ so that

$$\mu(A) < \delta \implies \nu(A) < \epsilon$$

Proof: ($\Leftarrow$) If $\mu(A) = 0$, then $\mu(A) < \delta$ for any δ, so $\nu(A) < \epsilon$ for any ϵ, so $\nu(A) = 0$.

($\Rightarrow$) Suppose not. Then, for some $\epsilon > 0$ and every k, there exists A_k such that $\mu(A_k) < 2^{-k}$ but $\nu(A_k) > \epsilon$. Let

$$B = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

Then,

$$\mu(B) = \lim_{n\to\infty} \mu\left(\bigcup_{k=n}^{\infty} A_k\right) \leq \lim_{n\to\infty} \sum_{k=n}^{\infty} 2^{-k} = 0$$

Now, $\mu(B) = 0$, but as ν is finite,

$$\nu(B) = \lim_{n\to\infty} \nu\left(\bigcup_{k=n}^{\infty} A_k\right) \geq \epsilon$$

This contradicts absolute continuity. Note that we used the finiteness of ν to make sure these descending limits exist.

■

Theorem 21.4: Radon-Nikodym Theorem

Let μ be σ-finite and ν be a finite measure. Suppose $\nu \ll \mu$. Then, there is an integrable $f \geq 0$ so that

$$\nu(A) = \int_A f d\mu$$

and f is μ-a.e. unique. We call f the *Radon-Nikodym derivative* of ν, and denote it $\frac{d\nu}{d\mu}$.

Proof: We need a quick lemma.

Lemma 21.5

Let μ and ν be finite measures. If μ and ν are not mutually singular, then there is some $\epsilon > 0$ and some set A so that $\mu(A) > 0$ and A is a positive set for $\nu - \epsilon\mu$. That is, $\nu(B) \geq \epsilon\mu(B)$ for $B \subseteq A$.

Proof: Apply the Hahn decomposition to $\nu - \frac{1}{n}\mu$ to get positive sets F_n and negative sets E_n. Let $E = \bigcap_n E_n$. For each n,

$$\nu(E) \leq \nu(E_n) \leq \frac{1}{n}\mu(E_n) \leq \frac{1}{n}\mu(X)$$

Taking $n \to \infty$, we see that $\nu(E) = 0$.

Consider $\mu(E^c)$. We can't have $\mu(E^c) = 0$, as this would imply that μ and ν are mutually singular. Thus, $\mu(E^c) > 0$. We see $E^c = \bigcup_n E_n^c$, so for some n, $\mu(E_n^c) > 0$. Let $A = E_n^c = F_n$ and let $\epsilon = \frac{1}{n}$. Then, $\nu - \frac{1}{n}\mu$ is positive on A and $\mu(A) > 0$.

■

Proof of Theorem 21.4 (Radon-Nikodym): Start by assuming μ is finite. Let

$$F = \left\{ g : X \to [0, \infty) \ : \ \int_A g d\mu \leq \nu(A) \ \forall A \in \mathcal{A} \right\}$$

F is nonempty because it contains 0.

Now, let $L = \sup\{\int g d\mu \ : \ g \in F\}$. Choose f_n so that $\int f_n d\mu \to L$, and let $g_n = f_1 \vee \cdots \vee f_n$. We want to show that $g_n \in F$. If $h, g \in F$, then

$$\begin{aligned}
\int_A (h \vee g) d\mu &= \int_{A \cap \{h \geq g\}} (h \vee g) d\mu + \int_{A \cap \{h < g\}} (h \vee g) d\mu \\
&\leq \int_{A \cap \{h \geq g\}} h d\mu + \int_{A \cap \{h < g\}} g d\mu \\
&\leq \nu(A \cap \{h \geq g\}) + \nu(A \cap \{h < g\}) \\
&= \nu(A)
\end{aligned}$$

so $h \vee g \in F$. Thus, by induction, $g_n \in F$ for all n. Now, let $f = \lim_{n \to \infty} g_n$. By the MCT,

$$\int_A f d\mu = \lim_{n \to \infty} \int_A g_n d\mu \leq \nu(A)$$

so $f \in F$. Also, $\int_X f d\mu = L$, by the MCT.

Now, we want to show that $\nu(A) = \int_A f d\mu$. Let

$$\tilde{\nu}(A) = \nu(A) - \int_A f d\mu$$

We see that $\tilde{\nu}$ is a measure, so we want to show it is zero.

We know $\tilde{\nu} \ll \mu$, so we want to show $\tilde{\nu} \perp \mu$. Suppose not. By our lemma, we can find A and ϵ so that A is positive for $\tilde{\nu} - \epsilon\mu$ and $\mu(A) > 0$. Then, for $B \subseteq A$,

$$\nu(B) \geq \int_B f d\mu + \epsilon\mu(B)$$

Because of this, we know that $f + \epsilon\chi_A \in F$, as

$$\begin{aligned}
\int_B (f + \epsilon\chi_A) d\mu &= \int_B f d\mu + \epsilon\mu(A \cap B) \\
&\leq \int_B f d\mu + \nu(A \cap B) - \int_{A \cap B} f d\mu \\
&= \int_{B \setminus A} f d\mu + \nu(A \cap B) \\
&\leq \nu(B \setminus A) + \nu(A \cap B) = \nu(B)
\end{aligned}$$

However,

$$\int (f + \epsilon\chi_A) d\mu = \int f d\mu + \epsilon\mu(A) = L + \epsilon\mu(A) > L$$

This contradicts the maximality of L. Thus, $\tilde{\nu} \perp \mu$, so $\tilde{\nu} = 0$ by Proposition 21.2. Thus,

$$\nu(A) = \int_A f d\mu$$

To show uniqueness, suppose $\nu(A) = \int_A g d\mu$. Then,

$$\int_A (f - g) d\mu = 0$$

for all A. Then, by a previous result (Proposition 13.4), we know $f - g = 0$ a.e., so $f = g$ a.e.

To extend to σ-finite μ, write $X = \bigsqcup_n X_n$ so that each X_n has finite measure. Then, apply the above to $\mu\,|_{X_n}$. Thus, we get a function f_n on each X_n with $\nu(A) = \int_A f_n d\mu$ for $A \subseteq X_n$. Now, defining $f = \sum_n f_n \chi_{X_n}$, we have that

$$\nu(A) = \int_A f d\mu$$

for any A by countable additivity and the MCT, integrability holds as ν is finite, and uniqueness one ach X_n extends to uniqueness everywhere.

■

10/19/26 Lecture 22: Lebesgue Decomposition and Differentiation

Theorem 22.1: Lebesgue Decomposition Theorem

Let μ be positive and σ-finite and ν positive and finite. Then, there exists ρ positive satisfying $\rho \perp \mu$ and $f \geq 0$ integrable such that

$$\nu(A) = \int_A f d\mu + \rho(A)$$

Proof: Assume μ is finite and let f be as in the proof of the Radon-Nikodym theorem; that is, such that $\int f d\mu = L = \sup\{\int g d\mu \, : \, g \in F\}$, where

$$F = \left\{ g : X \to [0, \infty) \, : \, \int_A g \, d\mu \leq \nu(A) \; \forall A \in \mathcal{A} \right\}$$

Define $\rho(A) = \nu(A) - \int_A f d\mu$. We see that $\rho(A)$ is positive from the fact $f \in F$.

We want $\rho \perp \mu$. If not, by Lemma 21.5 we can find a set A and $\epsilon > 0$ such that $\mu(A) > 0$ and A is positive for $\rho - \epsilon\mu$. Then, for $B \subseteq A$,

$$\epsilon\mu(B) \leq \nu(B) - \int_B f d\mu$$

Now, consider $f + \epsilon\chi_A$. Given B, we see that

$$\begin{aligned} \int_B f + \epsilon\chi_A d\mu &= \int_{B \setminus A} f d\mu + \int_{B \cap A} f + \epsilon\chi_A d\mu \\ &\leq \nu(B \setminus A) + \nu(B \cap A) \\ &= \nu(B) \end{aligned}$$

where for the second term we used that $\nu(B) \geq \int_B f d\mu + \epsilon\mu(B)$ for $B \subseteq A$.

Thus, $f + \epsilon\chi_A \in F$, which contradicts that $\int f d\mu = L$ is the sup over integrals in F. Thus, we must have $\rho \perp \mu$, which was to be shown.

■

Example: Necessity of Absolute Continuity

We need absolute continuity to have R-N derivatives. For instance, if we wanted $\frac{d\delta_0}{d\mu}$, where μ is the Lebesgue measure, we would want an f such that

$$\delta_0(\{0\}) = \int_{\{0\}} f d\mu = 1$$

but no f exists since $\{0\}$ has measure 0.

Corollary 22.2

Let ν be a finite measure and μ σ-finite with $\nu \ll \mu$. If f is integrable with respect to ν, then $f\frac{d\nu}{d\mu}$ integrable also with respect to μ with

$$\int f d\nu = \int f \frac{d\nu}{d\mu} d\mu$$

Proof: For a simple function χ_A,

$$\int \chi_A \frac{d\nu}{d\mu} d\mu = \int_A \frac{d\nu}{d\mu} d\mu = \nu(A) = \int \chi_A d\nu$$

By linearity, we get simple functions and by monotone convergence, we get non-negative and integrable functions.

■

Definition 22.3: Lipschitz Functions

$f : \mathbb{R} \to \mathbb{R}$ is *Lipschitz* if there exists L such that $|f(x) - f(y)| \leq L\,|x - y|$ for $x, y \in \mathbb{R}$.

Example: C^1 Functions

If $f \in C^1$ and $|f'| \leq L$, then f is L-Lipschitz. The converse is not true. For example, $|x|$ is 1-Lipschitz.

Proposition 22.4

Let $f : \mathbb{R} \to \mathbb{R}$ be Lipschitz. Then, f is differentiable a.e.

Proof: We will soon prove the following fact that we will use here: If $f(x) = \int_a^x g(y)dy$, then f is differentiable a.e. and $f' = g$.

Let L be such that $|f(x) - f(y)| \leq L\,|x - y|$. First, suppose f is increasing and let m be its L-S measure. Then

$$\begin{aligned} m((a, b]) &= f(b) - f(a) \\ &\leq L(b - a) \end{aligned}$$

This implies $m \ll \lambda$. Then, $m(A) = \int_A g(y)dy$ for some g. Thus,

$$m((a, x]) = \int_a^x g(y)dy = f(x) - f(a)$$

so $f(x) - f(a)$ is differentiable a.e.

If f is not increasing, then $f + Lx$ must be. This is because if $x > y$,

$$\begin{aligned} (f(x) + Lx) - (f(y) + Ly) &= (f(x) - f(y)) + L(x - y) \\ &\geq 0 \quad \text{as } f(x) - f(y) \geq -L(x - y) \end{aligned}$$

Thus, we see $f + Lx$ is differentiable a.e. by above, so f must be as well. ■

Differentiation

In this section, we want to show that the derivative of $\int_a^x f(y)d\mu(y)$ is equal to $f(x)$ μ-a.e. Even stronger, we want to study

$$f_r(x) = \frac{1}{\lambda(B_r(x))}\int_{B_r(x)} f(y)dy$$

where $B_r(x) = \{y : d(x,y) < r\}$.

Q: Does $f_r \to f$ as $r \to 0$? In 1d, we have

$$f_r = \frac{1}{2r}\int_{x-r}^{x+r} f(y)dy$$

First, suppose g is continuous. Then,

$$\begin{aligned}|g_r(x) - g(x)| &= \left|\frac{1}{\lambda(B_r(x))}\int_{B_r(x)} g(y) - g(x)dy\right| \\ &\leq \frac{1}{\lambda(B_r(x))}\int_{B_r(x)} |g(y) - g(x)|\,dy\end{aligned}$$

Since g is continuous, we see that for any $\epsilon > 0$, there exists r such that $y \in B_r(x) \implies |g(y) - g(x)| < \epsilon$. Then, for such r,

$$|g_r(x) - g(x)| \leq \frac{1}{\lambda(B_r(x))}\int_{B_r(x)} |g(y) - g(x)|\,dy < \epsilon$$

so taking $\epsilon \to 0$, we see $g_r(x) \to g(x)$. Thus, if g is continuous, we are done.

For general integrable f, we want to pick g that is continuous and compactly supported such that

$$\int |f - g|\,dy < \epsilon$$

Thus,

$$|f_r(x) - f(x)| \leq |f_r(x) - g_r(x)| + |g_r(x) - g(x)| + |g(x) - f(x)|$$

We want to study the measure of the set where $|f_r(x) - f(x)| > \frac{1}{n}$. Note that we can control $|g(x) - f(x)|$ by Chebyshev's and $|g_r(x) - g(x)|$ by above, so we really want to study $|f_r(x) - g_r(x)|$, which is at most

$$\frac{1}{\lambda(B_r(x))}\int_{B_r(x)} |f(y) - g(y)|\,dy$$

We want to show that this is large on a small set using the fact that $|f(y) - g(y)|$ is small.

Definition 22.5: Hardy-Littlewood Maximal Function

We define

$$M(h(x)) = \sup_{r>0}\frac{1}{\lambda(B_r(x))}\int_{B_r(x)} |h(y)|\,dy$$

We want to show that Mh is controlled by $\int |h|$.

10/21/26 Lecture 23: The Maximal Function and Lebesgue Differentiation

Recall: For f integrable, $f : \mathbb{R}^d \to \mathbb{R}$,

$$f_r(x) = \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} f(y)dy$$

Does $f_r(x) \to f(x)$ as $r \to 0$?

In general: No. Let $f(x) = \chi_C$ for C the Cantor set. Then $f_r(x) = 0$. So $f(x) \neq f_r(x)$, but note $f(x) = 0$ a.e.

Definition 23.1: Locally Integrable Functions

We say f measurable is *locally integrable* if $\int_K |f| < \infty$ for any compact set K, and we write $f \in L^1_{\text{loc}}$.

Theorem 23.2: Lebesgue Differentiation Theorem

If $f \in L^1_{\text{loc}}$, then $f_r(x) \to f(x)$ a.e. as $r \to 0$.

We will work toward proving this, but not yet.

Proposition 23.3

If $f \in L^1_{\text{loc}}$, then $Mf(x)$ is measurable.

Proof: Given sets A and B,

$$\begin{aligned}
\left|\int_A f - \int_B f\right| &= \left|\int_A f - \left(\int_{A\cap B} f + \int_{B\setminus A} f\right)\right| \\
&= \left|\int_{A\setminus B} f - \int_{B\setminus A} f\right| \\
&\leq \int_{A\setminus B} |f| + \int_{B\setminus A} |f| \\
&= \int_{A\triangle B} |f|
\end{aligned}$$

Thus, for r fixed,

$$\left|\frac{1}{\lambda(B_r(x))} \int_{B_r(x)} f(y)dy - \frac{1}{\lambda(B_r(x'))} \int_{B_r(x')} f(y)dy\right| \leq \frac{1}{\lambda(B_r(0))} \int_{B_r(x)\triangle B_r(x')} |f|\, dy$$

As $x' \to x$, $\lambda(B_r(x)\triangle B_r(x')) \downarrow 0$, so the integral goes to 0 by DCT. Thus,

$$\frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |f|\, dy$$

is continuous in x. Thus, it is measurable, and as the sup of measurable functions is measurable, $Mf(x)$ is measurable.

■

Lemma 23.4: Wiener Covering Lemma

Let $\{B_\alpha\}_{\alpha\in I}$ be a collection of balls in $\mathbb{R}^d$ with $\lambda\left(\bigcup_\alpha B_\alpha\right) < \infty$. Then, for every $\epsilon > 0$, there exist finitely many disjoint $B_1, \dots, B_n$ such that

$$\sum_{j=1}^{n} \lambda(B_j) \geq 3^{-d}(1-\epsilon)\lambda\left(\bigcup_\alpha B_\alpha\right)$$

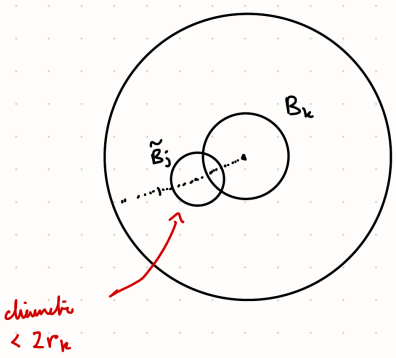
Proof: By inner regularity, we can find K compact so that $\lambda(K) > (1-\epsilon)\lambda\left(\bigcup_\alpha B_\alpha\right)$ and $K \subseteq \bigcup_\alpha B_\alpha$. Then, by compactness, we can find $\tilde{B}_1, \dots, \tilde{B}_m$ that cover K.

Let B_1 be the $\tilde{B}_j$ with the largest radius. Then, let B_j be the $\tilde{B}_j$ with the largest radius that does not intersect the $B_1, \dots, B_{j-1}$. Do this until we can no longer pick B_j. Let $B_1, \dots, B_n$ be our collection.

Now, for every $\tilde{B}_j$, we have that it is in our collection or intersects a ball in our collection. If it does intersect another ball in our collection, let B_k be the smallest index such that $\tilde{B}_j \cap B_k \neq \emptyset$. This implies $\tilde{B}_j$ does not intersect $B_1, \dots, B_{k-1}$.

Thus, we must have $\text{rad}(\tilde{B}_j) \leq \text{rad}(B_k)$ by our method of selection. Thus, writing $B_k = B_{r_k}(x)$ and $B_k^* = B_{3r_k}(x)$,

Claim: $\tilde{B}_j \subseteq B_k^*$. .



Thus, we see

$$(1-\epsilon)\lambda\left(\bigcup_\alpha B_\alpha\right) < \lambda(K) \leq \sum_{j=1}^{n} \lambda(B_j^*) = \sum_{j=1}^{n} 3^d \lambda(B_j)$$

which gives us the inequality.

■

Theorem 23.5: Weak 1-1 Inequality for M

There is a constant $C > 0$ so that for all $f \in L^1$ and any $\alpha > 0$,

$$\lambda(\{x : Mf > \alpha\}) \leq \frac{C}{\alpha} \int |f| \, dx$$

Proof: Let $A_\alpha = \{x : Mf > \alpha\}$. Then, for each $x \in A_\alpha$, we can find a ball $B_{r_x}(x)$ such that

$$\frac{1}{\lambda(B_{r_x}(x))} \int_{B_{r_x}(x)} |f| \, dy > \alpha$$

By our lemma, we can find finitely many disjoint balls $B_1, \ldots, B_n$ such that

$$3^{-d}(1-\epsilon)\lambda(A_\alpha) \leq \sum_{j=1}^{n} \lambda(B_j)$$

Thus,

$$\begin{aligned}(1-\epsilon)\lambda(A_\alpha) \leq 3^d \sum_{j=1}^{n} \lambda(B_j) &\leq \frac{3^d}{\alpha} \sum_{j=1}^{n} \int_{B_j} |f| \, dy \\ &\leq \frac{3^d}{\alpha} \int |f| \, dy\end{aligned}$$

because the B_j are disjoint. Then, taking $\epsilon \to 0$, we get

$$\lambda(A_\alpha) \leq \frac{3^d}{\alpha} \int |f| \, dy$$

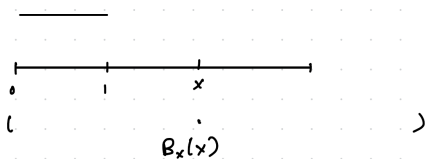
which was to be shown! ■

Remark: Suppose we knew $\int Mf dx \leq C \int |f| \, dx$; this is a strong 1-1 bound. Then, by Chebyshev,

$$\lambda\{Mf > \alpha\} \leq \frac{\int Mf}{\alpha} \leq \frac{C}{\alpha} \int |f|$$

which is our weak bound. However, this is typically not true.

Consider $d = 1$ and let $f = \chi_{[0,1]}(x)$.



then, we see that for $x > 1$,

$$\sup_{r>0} \frac{1}{2r} \int_{B_r(x)} \chi_{[0,1]}(y) dy \geq \frac{1}{2x}$$

so for $x > 1$, $Mf \geq \frac{1}{2x}$, which is not integrable. Thus, the strong 1-1 inequality can't hold.

Now, we go back to the theorem we want to prove.

Proof of Theorem 23.2: It suffices to consider $f\chi_{|x|\leq N}$, since we only care about the limit behavior of f_r. Thus, assume f is integrable. Given $\epsilon > 0$, find g continuous and compactly supported so that $\int |f - g| < \epsilon$. Then,

$$\begin{aligned}\limsup_{r\to 0} |f_r(x) - f(x)| &\leq \limsup_{r\to 0} |(f-g)_r(x) + g_r(x) - g(x) + g(x) - f(x)| \\ &\leq \limsup_{r\to 0} (|(f-g)_r(x)| + |g_r(x) - g(x)| + |g(x) - f(x)|) \\ &\leq M(f-g)(x) + 0 + \limsup_{r\to 0} |g(x) - f(x)|\end{aligned}$$

Let $E_\alpha = \{\limsup_{r\to 0} |f_r(x) - f(x)| > \alpha\}$ and $F_\alpha = \{|f(x) - g(x)| > \alpha\}$. Then,

$$E_\alpha \subseteq F_{\alpha/2} \cup \{M(f-g) > \alpha/2\}$$

We will finish next lecture using the fact that F_α is controlled by Chebyshev and $\{M(f-g) > \alpha/2\}$ is controlled by weak 1-1.

10/23/26 Lecture 24: Lebesgue Differentiation and Its Consequences

Theorem 24.1: Lebesgue Differentiation Theorem

If $f \in L^1_{\text{loc}}$, then $\lim_{r\to 0} f_r(x) = f(x)$ a.e.

Proof: Recall that it suffices to take $f \in L^1$. We see

$$\{x : f_r(x) \not\to f(x)\} \subseteq \bigcup_n A_n$$

where $A_n = \{x : \limsup_{r\to 0} |f_r(x) - f(x)| > 1/n\}$. We want to show $\lambda(A_n) = 0$ for all n. Given $\epsilon > 0$, there exists g continuous and integrable such that

$$\int |f - g|\, dx < \epsilon$$

Then,

$$\begin{aligned}\limsup_{r\to 0} |f_r(x) - f(x)| &\le \limsup_{r\to 0} \left(|f_r(x) - g_r(x)| + |g_r(x) - g(x)| + |g(x) - f(x)|\right) \\ &\le M|f - g|(x) + 0 + |f(x) - g(x)|\end{aligned}$$

Then,

$$A_n \subseteq \left\{x : M|f - g|(x) > \frac{1}{2n}\right\} \cup \left\{x : |f(x) - g(x)| > \frac{1}{2n}\right\}$$

By the weak 1-1 inequality,

$$\lambda\left\{x : M|f - g|(x) > \frac{1}{2n}\right\} \le \frac{C}{1/(2n)} \int |f - g|\, dx \le 2nC\epsilon$$

and by Chebyshev,

$$\lambda\left\{x : |f(x) - g(x)| > \frac{1}{2n}\right\} \le \frac{1}{1/(2n)} \int |f - g|\, dx \le 2n\epsilon$$

Then, as ϵ is independent of n and C, we see that by taking $\epsilon \to 0$, $\lambda(A_n) = 0$. ■

Now, we derive some interesting theorems.

Recall that, from the homework, if $A \subseteq \mathbb{R}$ such that $\lambda(A) > 0$, then for all $\alpha \in (0, 1)$ there exists an interval I such that $\lambda(A \cap I) > \alpha\lambda(I)$. It turns out this comes for free!

Corollary 24.2: Lebesgue Density Theorem

Suppose $A \subseteq \mathbb{R}$ is measurable such that $\lambda(A) > 0$. Then, for a.e. $x \in A$,

$$\lim_{r \to 0} \frac{\lambda(A \cap B_r(x))}{\lambda(B_r(x))} = 1$$

and for a.e. $x \notin A$,

$$\lim_{r \to 0} \frac{\lambda(A \cap B_r(x))}{\lambda(B_r(x))} = 0$$

Proof: Apply the theorem to $f = \chi_A$, as

$$\int_{B_r(x)} \chi_A = \lambda(A \cap B_r(x))$$

■

From our theorem, we also see

$$\lim_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} f(y) - f(x) dy = 0$$

Proposition 24.3

If $f \in L^1_{\text{loc}}$, then

$$\lim_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| \, dy = 0 \quad \text{a.e. } x$$

Proof: For $q \in \mathbb{Q}$, apply the theorem to $|f(x) - q|$ to see

$$\frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |f(y) - q| \, dy \to |f(x) - q|$$

for all $x \notin A_q$ (for a null set A_q).

If $x \notin A_q$ for any q, choose q_* so that $|f(x) - q_*| < \epsilon$. Then,

$$\begin{aligned} \limsup_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| \, dy &\leq \limsup_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |f(y) - q_*| \, dy \\ &\quad + \limsup_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} |q_* - f(x)| \, dy \\ &\leq 2\epsilon \end{aligned}$$

■

Corollary 24.4

Let $f : \mathbb{R} \to \mathbb{R}$ be integrable and fix some $a \in \mathbb{R}$. Then, let

$$F(x) = \int_a^x f(y)dy$$

then F is differentiable almost everywhere and $F'(x) = f(x)$ a.e.

Proof:

Let $h > 0$. Then,

$$F(x+h) - F(x) = \int_x^{x+h} f(y)dy$$

so

$$\begin{aligned}\left|\frac{F(x+h)-F(x)}{h} - f(x)\right| &= \left|\frac{1}{h}\int_x^{x+h} f(y) - f(x)dy\right| \\ &\leq \frac{2}{\lambda(B_h(x))}\int_{x-h}^{x+h} |f(y) - f(x)|\, dy\end{aligned}$$

Then, as $h \to 0$, the RHS goes to 0, so

$$\lim_{h\to 0} \frac{F(x+h)-F(x)}{h} = f(x)$$

almost everywhere.

■

Definition 24.5: Regular Borel Measures

We say that a Borel measure ν on $\mathbb{R}^d$ is *regular* if

1. $\nu(K) < \infty$ for any compact set K.
2. $\nu(A) = \inf\{\nu(\mathcal{U}) : \mathcal{U} \text{ open and } A \subseteq \mathcal{U}\}$.

Example: Regular Measures

Counting measure is not regular! Lebesgue and Lebesgue-Stieltjes measures are.

Proposition 24.6

Let ν be a finite positive regular Borel measure. Let

$$\nu(A) = \int_A f d\lambda + \rho(A)$$

be its Lebesgue decomposition with respect to Lebesgue measure. Then, for λ-a.e. $x \in \mathbb{R}^d$,

$$\lim_{r \to 0} \frac{\nu(B_r(x))}{\lambda(B_r(x))} = f(x)$$

Proof: By Lebesgue differentiation,

$$\lim_{r \to 0} \frac{1}{\lambda(B_r(x))} \int_{B_r(x)} f d\lambda = f(x) \quad \lambda\text{-a.e.}$$

Now, we want to show

$$\lim_{r \to 0} \frac{\rho(B_r(x))}{\lambda(B_r(x))} = 0$$

Choose A so that $\rho(A) = \lambda(A^c) = 0$. Notice that A^c is a λ-null set, so we just need to show that this limit holds on A.

Let $A_n = \left\{ x \in A : \limsup_{r \to 0} \frac{\rho(B_r(x))}{\lambda(B_r(x))} > 1/n \right\}$. We want to show $\lambda(A_n) = 0$ for all n. By regularity, find $\mathcal{U} \supseteq A$ (A is a ρ-null set) such that

$$\rho(\mathcal{U}) < \epsilon$$

If $x \in A_n$, we can find $B_{r_x}(x) \subseteq \mathcal{U}$ so that $\rho(B_{r_x}(x)) > \frac{1}{n} \lambda(B_{r_x}(x))$. Applying the Wiener covering lemma to $\bigcup_x B_{r_x}(x)$, we get disjoint $B_1, \dots, B_N$ such that

$$\begin{aligned}
(1 - \delta) \lambda \left(\bigcup_x B_{r_x}(x) \right) &\leq 3^d \sum_{j=1}^N \lambda(B_j) \\
&\leq 3^d \sum_{j=1}^N n \rho(B_j) \\
&\leq 3^d n \rho(\mathcal{U}) \\
&\leq 3^d n \epsilon
\end{aligned}$$

Taking $\delta = \frac{1}{2}$ and $\epsilon \to 0$, we see $\lambda\left(\bigcup_x B_{r_x}(x)\right) \to 0$, so $\lambda(A_n) = 0$.

■

10/26/26 Lecture 25: Differentiating Monotone Functions and Bounded Variation

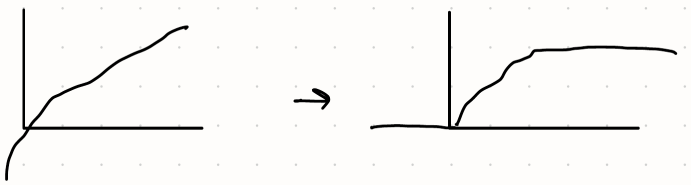
Proposition 25.1

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous. Then, $F'(x)$ exists a.e. and

$$\int_a^b F'(x)dx \leq F(b) - F(a)$$

Proof: Because differentiation is local, it suffices to prove the claim for a function that agrees with F on $[0, 1]$.

Redefine F so that $F(x) = \lim_{x \to 0^+} F(x)$ for $x \leq 0$ and $F(x) = F(1)$ for $x \geq 1$. Then,



F is still increasing and right continuous, but its LS measure is finite. Let ν be the LS measure of this F. Then,

$$\nu(A) = \int_A f d\lambda + \rho(A) \quad \text{for } \rho \perp \lambda$$

Then,

$$\begin{aligned}\rho((a, b]) &= \nu((a, b]) - \int_a^b f(y)dy \\ &= F(b) - F(a) - \int_a^b f(y)dy\end{aligned}$$

Now, ρ is a Lebesgue-Stieltjes measure, and

$$\frac{F(x + h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(y)dy + \frac{\rho((x, x + h])}{h}$$

so by the corollary from before, the first term goes to f almost everywhere, and by a proposition from last time,

$$\frac{\rho((x, x + h])}{h} \leq \frac{\rho(x - 2h, x + 2h)}{h} = 4\frac{\rho(x - 2h, x + 2h)}{\lambda(x - 2h, x + 2h)} \to 0$$

because ρ is mutually singular with λ. Thus,

$$\int_a^b F'(x)dx = \int_a^b f(x)dx \leq \nu((a, b]) = F(b) - F(a)$$

■

Example: Cantor Function

Let F be the Cantor function. F has the property that it is constant outside the Cantor set, so $F'(x) = 0$ a.e. Thus, the L-S measure of F is mutually singular with λ. Thus, we see

$$\int_a^b F'(x)dx = 0 \leq F(b) - F(a)$$

Thus, we don't always achieve equality.

Note: In our proof above, we see $\int_a^b F'(x)dx = F(b) - F(a)$ when the LS measure ν is absolutely continuous with respect to λ.

Theorem 25.2

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and let $G(x) = F(x^+) = \lim_{y \to x^+} F(y)$. Then, the set of points where F is not continuous is countable, and F and G are differentiable with

$$F' = G' \text{ a.e.} \quad \text{and} \quad \int_a^b F'(y)dy \leq F(b) - F(a)$$

Proof: G is increasing and right continuous.

Now, the points where F is discontinuous are points such that $F(x^+) \neq F(x^-)$ (and namely $F(x^-) < F(x^+)$).

At any such point, we can find a rational in $(F(x^-), F(x^+))$, but as F is increasing, all such intervals are disjoint, and thus, all such rationals are unique. Thus, the number of points F is discontinuous is countable.

Now, note that G agrees with F where F is continuous. Since G' exists a.e. from above, we can find x such that $G'(x)$ exists, $F(x) = G(x)$, and F is continuous at x.

Fix $\epsilon > 0$. For any $h \geq 0$,

$$F(x + h) \leq G(x + (1 + \epsilon)h)$$

This is because in $(x+h, x+(1+\epsilon)h)$, we can find a point x_h where F and G agree, and $G(x+(1+\epsilon)h) \geq G(x_h)$. Thus,

$$\begin{aligned}
\limsup_{h \to 0^+} \frac{F(x + h) - F(x)}{h} &\leq \limsup_{h \to 0^+} \frac{G(x + (1 + \epsilon)h) - G(x)}{h} \\
&= (1 + \epsilon) \limsup_{h \to 0^+} \frac{G(x + (1 + \epsilon)h) - G(x)}{(1 + \epsilon)h} \\
&= (1 + \epsilon)G'(x)
\end{aligned}$$

Thus, $\limsup_{h \to 0^+} \frac{F(x+h)-F(x)}{h} \leq (1 + \epsilon)G'(x)$. By a similar argument, $\liminf_{h \to 0^+} \frac{F(x+h)-F(x)}{h} \geq (1 - \epsilon)G'(x)$. Then, taking $\epsilon \to 0$, we get $F'(x) = G'(x)$.

Finally, to show $\int_a^b F'(x)dx \leq F(b) - F(a)$, given $a < b$, choose $a_n \downarrow a$ and $b_n \uparrow b$ so that F and G agree at

these points. Then,

$$\begin{aligned}\int_{a_n}^{b_n} F'(x)dx &= \int_{a_n}^{b_n} G'(x)dx \\ &\leq G(b_n) - G(a_n) \\ &= F(b_n) - F(a_n) \\ &\leq F(b) - F(a)\end{aligned}$$

where we apply MCT to get the inequality as F' is non-negative almost everywhere. ■

Let $F : \mathbb{R} \to \mathbb{R}$. Define

$$T_F(x) = \sup\left\{\sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| \ : \ n \in \mathbb{N},\ x_0 < x_1 < \cdots < x_n = x\right\}$$

If F is smooth, $T_F(x) = \int_{-\infty}^{x} |F'(t)|\, dt$.

Definition 25.3: Bounded Variation

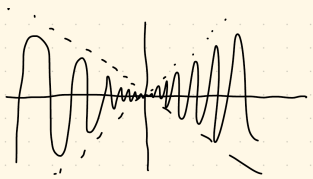
We say F is of *bounded variation*, denoted BV, if $\lim_{x\to\infty} T_F(x) = T_F(\infty) < \infty$.

Example: Bounded Increasing Functions

If F is bounded and increasing, then F is BV.

Example: $x\sin(1/x)$

Monotonicity is important. Consider $F(x) = x\sin\left(\frac{1}{x}\right)$.



This function is bounded near 0, but not BV on any interval containing 0.

10/28/26 Lecture 26: Functions of Bounded Variation

Recall: If $F : \mathbb{R} \to \mathbb{R}$ is increasing, F is differentiable a.e.

For $F : \mathbb{R} \to \mathbb{R}$, consider

$$T_F(x) = \sup \left\{ \sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| \ : \ n \in \mathbb{N}, \ x_0 < x_1 < \cdots < x_n = x \right\}$$

and if $T_F(\infty) < \infty$, $F \in \text{BV}$.

Now, if $x_{j-1} < x' < x_j$,

$$|F(x_j) - F(x_{j-1})| \leq |F(x_j) - F(x')| + |F(x') - F(x_{j-1})|$$

and if $a < b$,

$$T_F(b) = T_F(a) + \sup \left\{ \sum |F(x_j) - F(x_{j-1})| \ : \ a = x_0 < \cdots < x_n = b \right\}$$

Definition 26.1: Bounded Variation on an Interval

We say F is of *bounded variation on* $[a, b]$ if

$$T_F(b) - T_F(a) < \infty$$

and we write $F \in \text{BV}([a, b])$.

Proposition 26.2

If $F \in \text{BV}$, then $T_F + F$ and $T_F - F$ are increasing.

Proof: Fix $x < y$. By definition, choose $x_0 < \cdots < x_n = x$ so that $\sum_j |F(x_j) - F(x_{j-1})| \geq T_F(x) - \epsilon$. Then,

$$\begin{aligned} T_F(y) + F(y) &\geq \sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| + |F(y) - F(x)| + F(y) \\ &\geq T_F(x) - \epsilon + F(x) \end{aligned}$$

so taking $\epsilon \to 0$, we get what we want. A similar argument works for $T_F - F$. ■

Theorem 26.3: Classification of BV

A function $F : \mathbb{R} \to \mathbb{R}$ is BV iff it may be written as a difference of bounded increasing functions. In particular, $F \in \text{BV} \implies F$ is differentiable almost everywhere and F is bounded.

Proof: ($\Rightarrow$) By the proposition, we can write

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$$

and both are increasing. To show boundedness, we see that, for $x < y$,

$$F(x) - F(y) \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

and similarly

$$F(y) - F(x) \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

so we see F is bounded, so $T_F + F$ and $T_F - F$ are bounded.

($\Leftarrow$) Bounded increasing functions have bounded variation, and as $T_{H-G} \leq T_H + T_G$ by the triangle inequality, we see that the difference also has bounded variation.

■

Recall if F is increasing and right-continuous, we get the LS measure m_F by $m_F((a, b]) = F(b) - F(a)$.

Proposition 26.4

Let m be a positive Borel measure with the property that $m(A) < \infty$ if A is bounded. Then, if

$$F(x) = \begin{cases} m((0, x]) & x > 0 \\ 0 & x = 0 \\ -m((x, 0]) & x < 0 \end{cases}$$

then F is increasing, right continuous, and $m = m_F$.

Proof:
Increasing follows from $A \subseteq B \implies m(A) \leq m(B)$, and right continuity comes from

$$m((0, x]) = \lim_{h \to 0^+} m((0, x + h])$$

It is then clear that $m = m_F$.

■

Corollary 26.5

If m is a Borel measure with $m(A) < \infty$ for bounded A, then m is regular!

This gives us a correspondence $m_F \longleftrightarrow F$, but not quite, as if $G = F + C$ for a constant C, then $m_F = m_G$.

To fix this, we will normalize as follows: let

$$\text{NBV} = \{F \in \text{BV} : F \text{ is right continuous and } F(-\infty) = 0\}$$

Theorem 26.6: NBV Functions and Signed Borel Measures

If μ is a finite signed measure on $\mathbb{R}$, then $F(x) = \mu((-\infty, x])$ is NBV. Conversely, if $F \in \text{NBV}$, then there is a unique signed Borel measure μ with $\mu((a,b]) = F(b) - F(a)$.

Before we prove this, we need a lemma.

Lemma 26.7

If $F \in \text{BV}$ and is right-continuous, then so is $T_F(x)$.

Proof: Fix x and let $\alpha = T_F(x^+) - T_F(x)$. We want to show $\alpha = 0$. Given $\epsilon > 0$, choose δ so that

$$|F(x+h) - F(x)| < \epsilon \quad \text{and} \quad T_F(x+h) - T_F(x^+) < \epsilon$$

for $0 < h < \delta$ by right-continuity and the definition of limits. Find a partition $x = x_0 < \cdots < x_n = x + h$ so that

$$\begin{aligned}\sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| &\geq \frac{3}{4} |T_F(x+h) - T_F(x)| \\ &\geq \frac{3}{4}\alpha\end{aligned}$$

$$\sum_{j=2}^{n} |F(x_j) - F(x_{j-1})| \geq \frac{3}{4}\alpha - |F(x_1) - F(x)| \geq \frac{3}{4}\alpha - \epsilon$$

Now, find $x = y_0 < \cdots < y_m = x_1$ so that

$$\sum_{j=1}^{m} |F(y_j) - F(y_{j-1})| \geq \frac{3}{4} |T_F(x_1) - T_F(x)| \geq \frac{3}{4}\alpha$$

Thus,

$$\sum_{j=1}^{m} |F(y_j) - F(y_{j-1})| + \sum_{j=2}^{n} |F(x_j) - F(x_{j-1})| \geq \frac{3}{2}\alpha - \epsilon$$

But as this is just a partition (of $[x, x+h]$),

$$\begin{aligned}\sum_{j=1}^{m} |F(y_j) - F(y_{j-1})| + \sum_{j=2}^{n} |F(x_j) - F(x_{j-1})| &\leq T_F(x+h) - T_F(x) \\ &= (T_F(x+h) - T_F(x^+)) + (T_F(x^+) - T_F(x)) \\ &< \epsilon + \alpha\end{aligned}$$

so

$$\frac{3}{2}\alpha - \epsilon \leq T_F(x+h) - T_F(x) < \epsilon + \alpha$$

so taking $\epsilon \to 0$, we must see $\alpha = 0$. Since $T_F(x)$ is increasing and bounded, $T_F \in \text{BV}$. ■

Proof of Theorem 26.6: ($\Rightarrow$) If μ is a finite signed Borel measure, write $\mu = \mu^+ - \mu^-$ and let

$$F^{\pm}(x) = \mu^{\pm}((-\infty, x])$$

Each of these are BV and $F^{\pm}(-\infty) = 0$ by continuity. They are also right continuous by continuity.

($\Leftarrow$) Write $F = F^+ - F^- = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$. $F^{\pm}$ are both increasing and right continuous from above, so we get the LS measures m_{F^+} and m_{F^-}, and it can be checked that

$$m = m_{F^+} - m_{F^-}$$

has the property we want.

■